

Table 2. Protection systems

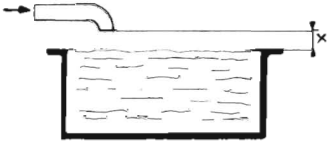
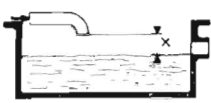
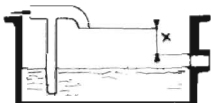
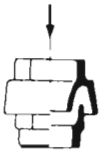
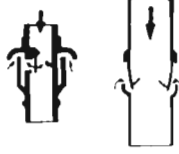
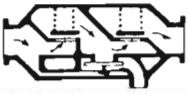
Method	Geometry	Remarks
Barometric loop	10 mm (3/32 in) loop without branch	Physically very safe, but costly and not always realisable. Inoperable in case of leakage.
Overflow safety gap:		
1. Total overflow		Essentially safe but ineffective if tap is extended with a sample pipe.
2. A partial or overflow limit		Safe if outlet flow is well dimensioned in comparison with inlet flow. Ineffectual if the tap is extended with a sample pipe.
3. Diverted overflow		Safe principle but limited in practical application.
Disconnectors:		
1. Without moving parts		Safe principle but may involve substantial head loss.
2. With moving parts		
(a) On-line		Safe principle, may have reliability problems.
(b) Tee		Safe principle, may have reliability problems.
Reduced pressure backflow preventer		Elaborate and costly system, generally needing regular maintenance to ensure continued proper function.
Non-return valves		Simple and effective. Performance and reliability primarily depend on design and quality of components and manufacture.

Table 3. State of the regulations in force in different countries

Country	Regulations with tests, agreements and/or standards	Control at the manufacturer's site	Control with the consumer or in the trade
Germany	X	—	—
Great Britain	X	—	X
Belgium	X	—	X
Denmark	X	X	X
Spain	—	—	—
France	X	X	X
Holland	X	X	X
Italy	—	—	—
Sweden	X	X	X
Switzerland	X	—	—

Back-flow preventers and vacuum breakers

Back-flow arising from connections between a potable and non-potable water supply can constitute a serious public health hazard. There are numerous well documented cases where back-flow from cross connections has resulted in contamination which could be harmful to health. The problem is a dynamic one because plumbing systems are continually being installed, altered and extended.

Where pollution cannot be prevented using approved check valves, the modern approach is to use vacuum breakers and back-flow preventers. Vacuum breakers are used on medium-risk lines and fawcets. In high risk areas, many Health Authorities throughout the world recommend the use of reduced-pressure back-flow preventers. These units operate on the variation of three decreasing pressures created by the head loss of two check valves. Any reversal of pressure from back-flow and/or back-siphonage is sensed by a membrane actuating a release valve which will drain polluted water to atmosphere.

Non-return valve families

An NRV is an automatic appliance intended to prevent a fluid changing its flow direction.

Numerous classifications exist. Here we use that of the CEN which distinguishes between four families, based on the direction of the displacement of the closing system with regard to that of the flow of water.

Note that there are other NRV families but these are not for use in the sanitary field, notably the ball valve, particularly recommended for loaded liquids.

Family 1:

Linear displacement, not parallel.
Perpendicular-displacement valve with straight flap.
Oblique-displacement valve with oblique flap.

Family 2:

Angular displacement.
Lift valve.
Butterfly valve.

Family 3:

With deformable closing system.
Membrane valve.

Family 4:

With linear parallel displacement.
'Guided' valves.



Back-flow preventer.

Test specifications on non-return valves

These should embrace the following points: mechanical characteristics of the casing, tightness, head loss, and reliability.

In the field of sanitary protection, different countries have enforced test specifications and methods for valves smaller than 50 mm (2 in). We will examine here this category only and will mention the minimum and maximum requirements for each of the characteristics listed previously.



Stainless-steel mini-fittings used to join small-bore stainless-steel tubing.



A selection of copper and gunmetal fittings.

Mechanical characteristic of the casing

The resistance is defined:

- (i) To pressure: from 16 bar (230 lbf/in²) in water at ambient temperature to 25 bar (360 lbf/in²) at 95°C (203°F), (duration of test 2 to 5 min).
- (ii) To buckling and torsion: according to the sizes and the countries, the casings are subject to constraint in order to determine their resistance.

Certain countries require the NRVs to be of dismountable design with regard to the connection, with bosses for the following purposes:

- (a) To control the tightness of the NRV.
- (b) To drain the installation.
- (c) To disinfect the pipes.

Also in certain countries, the alloys are specified according to the nature of the water (e.g. the problem of dezincification).

It must be noted that the mechanical holding tests carried out on the casing should not modify the hydraulic characteristics of the NRV.

Tightness

- (i) Low-pressure tightness: all specifications prescribe that an NRV must be tight under a pressure of 30 mm (1.3 in) water column, applied during a total time of 3–10 min.
- (ii) High-pressure tightness: the valve must be tight under a pressure rising from 0 to 16 bar (0 to 230 lbf/in²) applied during a total time of 3 to 10 min.

Head loss (ΔP)

We know that the head loss is a loss of pressure induced by any plumbing appliance mounted on a pipe. The design of the appliance can reduce the head loss to a minimum. In the case of an NRV, the survey of the hydraulic profile of the internal parts determines the head loss.

It is usually expressed in feet or metres water column at a given flow. (It can also be expressed by a coefficient of ΔP , k , square function of the average speed of flow on twice the acceleration of the weight.) The specifications usually prescribe, for a given nominal diameter, the achievement of a minimum flow at different values of ΔP ranging from 0.5 to 10 m (1.6 to 32 ft) water column.

Reliability

Endurance tests are very important for an NRV which, insofar as being a mechanical system, is often tainted with unfavourable prejudice in comparison with purely hydraulic systems. Without modification of its principal characteristics (tightness, ΔP), the appliance can undergo up to 100,000 cycles of operation at a pressure varying from 6 to 16 bar (90 to 230 lbf/in²) (according to the country), one half in cold water and the other half in hot water (temperatures varying from 65 to 95°C (150 to 200°F) according to the country). This corresponds on average to a minimum longevity of 10 years.

Opening pressure

In principle, present legislation prescribes that an NRV must be able to function in all positions, which means in this case the use of a release spring. This spring must conform to the specifications which impose a limit of opening pressure. At present, the opening pressure must always be positive and below 1 m (3.28 ft) water column, according to the country. This is very important. Experience shows that, being open for 80% of its time, an NRV functions with very weak flow rates. Under these conditions, it is of prime necessity that the opening is complete, as vibrations occur which can generate strong noise nuisances, as well as their being problems of premature wear.

Valve types compared

At present, only the linear parallel-displacement valve described as 'guided' seems to comply as, whatever its position, the design of this type alone gives the following results:

- (i) Perfect tightness at weak pressure, mainly ensured by the quality of the guide system (difficult to obtain in the valve family with angular-displacement, lift or butterfly valves).
- (ii) A high-pressure tightness, mainly ensured by the compression of one face against the seat and a solid closing system, which is not always the case with the family of membrane valves.
- (iii) Minimum head loss, the linear parallel displacement preventing a considerable deformation of the fluid flow, contrary to that given by valve families producing non-parallel linear displacement.

Role and interest of the regulations

Regulations can only exist if they are defined and imposed by the public authorities and if the application is controlled by the proper bodies.

Regulations depend on two aspects:

- (i) The localisation of the protection features: it defines the list of appliances or installations in which a form of protection must be installed, in which part of the installation and what type of protection must be used.

Most European countries have prescribed their own regulations by replying to two questions: 'What is to be protected?' and 'In what manner?'. The result is that, in certain cases, the choice of appliance is left to the decision of the installer.

- (ii) The definition of the qualitative criteria of the protection equipment: From the moment when the fitter has chosen a type of appliance, it is necessary for him to know that the appliance corresponds with a qualitative specification which allows it to be used in the considered case.

Conformity to criteria

In all the countries where regulations exist, these criteria have been the object of tests, where the operating mode is defined and where the positive result is confirmed by the issue of an Approval No. and the right to affix a distinctive sign on the body of the valve.

The control of the appliance is either carried out by the body which has prescribed the regulations or by an authorised independent laboratory, whose results are confirmed by the recommended body.

It is desirable that the definition of the specification is the result of co-operation between the legislators, the manufacturers and the installers but it must not be forgotten that the opinion of the legislator is predominant.

Control of appliances when being designed and manufactured is insufficient if the complete installation itself is not also submitted to the control of the authorities. In the case of the whole installation having to be joined with an exterior source, it is the authorisation of the branch which is the ultimate sanction and constitutes its certificate of conformity.

Finally, it is necessary to watch that the quality of the installation does not deteriorate with time. This is the purpose of the necessary periodic controls, carried out by the responsible authorities and which particularly apply to the appliances subjected to greater approval requests. This is the reason why NRVs are equipped with bosses in order that controls can be carried out easily. In some cases the regulations even state quite simply that the appliances are to be exchanged every 5 years, hence the importance of the valves being of standard dimensions and also being easily dismountable.

Equally, it is ultimately necessary that regulations and specifications should extend beyond the national standards, so that the rules applicable to different countries are rendered identical for all. In the NRV field, the specialist commissions of the European Standards Committee are studying this problem, with the participation of the different national plumbing syndicates. It is

desired that their studies result in rationalisation, reduction of costs and the elimination of protectionism, of which some national regulatory bodies could presently be accused.

Pipes and tubes

The most common material used in domestic heating and plumbing systems is copper tube. This has proven to be a versatile and reliable tubing product, easily joined by compression fittings or more efficiently by soldering and brazing. Other methods for joining larger-diameter copper tubing include the grooved-end upper connection system that eliminates leaks more commonly associated with soldering and brazing and speeds up installation. The system is more suited to industrial applications and uses a pressure-responsive synthetic rubber gasket to seal on the outside of the tubing. Tube sizes range from 50 to 200 mm (2 to 8 in).

Increasing use is being made of stainless-steel tubing. Particular advantages offered by stainless steel tubes are:

- (a) The appearance of finished pipework is aesthetically pleasing. Both the satin and the polished finishes are attractive, and the thin walls of capillary fittings give pipeline and fittings a neat, continuous appearance.
- (b) No maintenance is required after installation; the satin finish stays satin-looking and the polished finish stays polished.
- (c) The price is comparatively stable, not fluctuating like copper.
- (d) The corrosion-resistance of stainless steel is better than copper in areas of cupro-solvency and it is not prone to pitting corrosion by water.
- (e) The mechanical properties are good. The strength is high which means that it is less prone to damage in service. Elongation is also high which gives it good bending properties—as a comparison, it takes the same amount of effort to bend 15 mm (0.6 in) stainless tube as a 22 mm (0.9 in) copper tube. It can be sawn easily with a hacksaw or roller cutter.
- (f) Copper or other non-ferrous pipe and fittings are subject to pilfering on site as they have a high scrap value. Stainless steel does not have a high second-hand value.

The inherent disadvantage of stainless-steel tubing employed as pressure pipes has been the difficulty of making leak-free joints (e.g. stainless-steel tubing used on high-pressure aircraft hydraulic systems is invariably welded). However, this has now largely been overcome by the availability of suitable fittings, solders and fluxes making plumbed joints quite practicable and several jointing techniques are now well established. These fall into two major categories:

- (i) techniques requiring heat
- (ii) joints which can be made at room temperature.

Joints using heat

Techniques using heat to join are welding, soft soldering and silver soldering. Welding is not used in the majority of plumbing installations although it is an established technique for the chemical, food and cryogenic industries. Soldering implies the use of capillary fittings where the i.d. of the fitting socket is just a few thous greater than the o.d. of the tubing. Capillary attraction pulls the solder into this gap. To make successful joints in stainless steel it is essential to recognise two important properties of the metal:

- (a) The low thermal conductivity of stainless steel—about $\frac{1}{30}$ th that of copper.
- (b) The property which gives the metal its stainlessness—this is because a hard oxide coating is formed within seconds of a nascent surface being presented to an oxygen-bearing atmosphere. The coating steadily increases in thickness until a stable protection is achieved.

For successful soft soldering, the outside end of the tube and the inside of the fitting socket must be abraded with emery cloth to remove the oxide skin. The prepared surface should then be painted with solder paste and the joint assembled.

The secret of soldering stainless steel well is not to hurry, to use a small flame and to make sure that the back of the joint (away from the plumber) gets enough heat. Using a metal reflector behind the joint can often be a help or, alternatively, the use of a cyclone burner produces a flame which will curl around the back of the fitting. When the solder begins to flow a frying sound comes from the joint. The joint should be completed by end-feeding with solder wire.

Many solder paints have been made and marketed over the years. One with a 5-year shelf life contains phosphoric acid base flux and also powder of the tin-lead alloy which has the lowest melting point possible. Phosphoric acid fluxes are recommended as they only become aggressive and eat up the tenacious chromium oxide when heated; they will not continue to attack the stainless steel when cold.

For silver soldering, it is necessary to use an aggressive chloride-based flux to eat the oxide and make a good joint. It is important to remember that this flux must be removed from the joint area (inside and outside) within 24 hours of making the joint. Usually swilling with water for an hour or so will remove all trace.

There are many suppliers who market silver solder and brazing rod, together with flux, especially for joining stainless steel. The majority of these are entirely satisfactory providing the suppliers' instructions are carefully followed, and there must be the same care in heating the joint as with soft soldering.

Capillary fittings are at their most efficient when the gap between tube and socket is uniform. Such uniformity is achieved with a design in which the tube is held centrally by a specific deformation of the socket at three points.

Joints made cold

Low conductivity has undoubtedly been the cause of a number of leaking joints made by soldering and, although soldering techniques have been much improved, it is still useful to explore cold techniques. These might be used, for instance, where naked flames cannot be applied, e.g. on sites with fire risks or pipelines fitted with a plastic anti-burst liner.

Three basic techniques are: cold adhesive bonding (now approved for cold-water supplies), the use of compression fittings, and the use of screw-thread fittings (the last named is only used for thick-walled pipe and, as it is not applicable to thin-walled tubing used in plumbing, will not be discussed further).

Experimentation has been carried out by the adhesive makers, by stainless-steel capillary-fitting makers and by the British Steel Corporation to determine the best adhesive to use and to devise a method of applying same. Of the many adhesives tested, the dimethacrylate esters, which are anaerobic adhesives, have proved to be the most satisfactory.

These anaerobic adhesives cure in the absence of air, that is, when they enter the confined capillary gap they start to harden. They will cure by themselves, but in order to make a joint cure to handling strength in 2 min it is necessary to use an accelerator which can be applied from an aerosol spray.

The bonding technique entails cleaning the tube ends with coarse emery cloth, spraying with accelerator and allowing the latter to dry (30 sec). A ring of the viscous adhesive is then applied to the leading edges of both the tube and the fittings. The tube is then pushed into the fitting socket and left without movement for 2 min for handling strength to be achieved. When cured, a pull-out strength of 1.1 tons is obtained for a 15 mm (0.6 in) tube. Pipelines jointed by these adhesives will withstand internal pressures of 310 bar (4500 lbf/in²).

This technique has now been approved by the UK Thames Water Authority for use with potable water and by the majority of the Regional Water Authorities in the UK. The maximum temperature for which its use with water and aqueous chemicals is recommended is 60°C (140°F).

Now that mini-bore tubing is available down to 6 mm (¹/₄ in), together with a complete range of low-cost capillary fittings, this technique (and soft soldering for applications above 60°C (150°F)) will increase the use of stainless-steel instrumentation, for medical gases and for various water applications.

Stainless-steel compression fittings employ the principle of tightening a nut which causes a ferrule or olive to be compressed tightly on the tube by forcing it down a cone on to the tube end. The shape of the ferrule is the subject of many patents.

Most compression fittings have two or three ferrules per fitting. Some designs dig into the tube wall—the so-called 'bite-ring' fittings—and some are just compressed tightly onto the tube. Joints are quickly and easily made with

compression fittings; with most designs just two spanners, turned in opposition, are needed. A particular model employs only one screw thread to apply the forces necessary for compressing the two ferrules and the joint can be broken and remade a number of times. These fittings are intended for use with thin-walled tube to BS4127, and this point should be checked by plumbers contemplating the purchase of stainless-steel compression fittings as some designs only work on thick-walled tube.

For the handling of some liquids such as milk, milk products and beer, it is essential to have demountable joints in order to clean stainless-steel pipelines. Such joints must be free from cracks and crevices which might become breeding places for bacteria. There are a number of designs of fittings available incorporating PTFE washers and neoprene O-rings for these applications. For other chemicals, cone-seated or flat-seated joints with gaskets are in common use.

Summary

The cause of leaks in stainless-steel plumbed joints in the past can be mainly attributed to lack of knowledge of the differences between stainless steel and copper. It is worth repeating that the low thermal conductivity of the metal must be recognised. A big flame from a blow-lamp is not enough; time must be allowed for the heat to soak all round a joint.

The other important difference is seen in the choice of flux. Chloride-based fluxes are a hazard as they are corrosive on stainless steel. Often hygroscopic traces of hydrochloric acid can form and cause pitting.

The availability of stainless plumbing as a complete system acts as a stabilising influence on the cost of plumbing. The growth of the stainless-steel domestic plumbing market will be affected by three main factors:

- (a) the price of copper tubing and fittings, which may well rise again in the not too distant future;
- (b) the price of stainless-steel compression fittings, which is expected to fall;
- (c) the availability of an adhesive which will withstand boiling water for long periods without losing its strength. When this is available, the use of stainless-steel tubing in central-heating systems will be even more widely accepted.

Domestic water-supply valves

The demand for shut-off and thermostatically-controlled valves has increased for domestic copper and stainless-steel pipelines with the greater use of domestic central heating, washing machines, dishwashers and showers, etc.

The quarter-turn valves shown in Figure 8 are designed as emergency shut-off valves to be installed upstream of taps, ball valves and appliances, and can be

used to isolate individual fittings for servicing without having to drain down the whole plumbing system. Alternatively, they are used for permanent plumbing-in of washing machines and other appliances.

The valve has been designed to shut against a test pressure of 20 bar (300 lbf/in²) and has been accepted by the UK National Water Council providing that it does not replace the mandatory screw down stop valve.

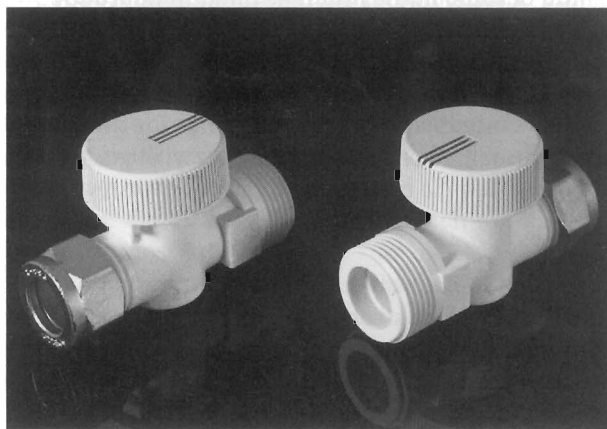


Figure 8. Quarter-turn shut-off valves.

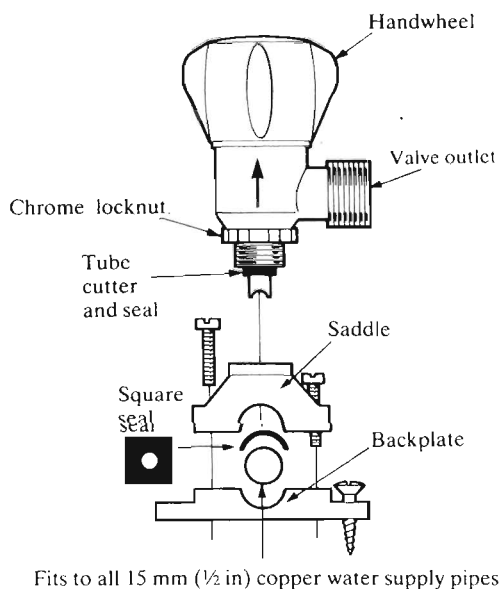


Figure 9. Self-cutting plumbing-in valve kit for domestic equipment.

A valve with a self-cutting plumbing-in kit has been developed so that it can be fitted quickly without having to turn off the water supply. The valve shown in Figure 9 is suitable for connection to 15 mm copper pipe supplying domestic cold and low-pressure hot water, and is ideal for plumbing-in washing machines, drinks dispensers and garden taps, etc.

Thermostatically-controlled flow regulators and stopcocks have been developed for use with electric instantaneous showers, wall kettles, etc., to enable a selected temperature to be maintained. The regulator senses fluctuations in supply pressure causing variations in water flows and automatically adjusts to provide constant flow across the heater elements, and thus provide constant temperature.

Hygienic Services

Valves have been developed for use in hygienic pipelines for dairy, brewery, food, beverage, biological and other process plants, where automatic valves manufactured to a high standard of hygienic design are demanded. These are usually stainless-steel valves free from pockets or crevices designed for the control of both product and 'cleaning-in-place' fluids.

Food and beverages

The food and beverage industry is a large user of gas, steam and water on a continuous basis, indicating a requirement for general process-control valves in addition to sterile valves for handling food products. Plastic/polymer valves are also a popular choice although use of these may be limited by the need for aggressive or caustic cleaning. Manual shut-off valves are used in areas of planned maintenance shut-down or in problem areas which may need to be cleared. Products can then be left in the pipeline leaving only a small section to be cleared and cleaned.

Double-seat valves are ideally suited for 'contained-flow systems', eliminating the need for manual swing bends for product and cleaning lines. Filling, emptying and CIP can take place simultaneously in a totally 'closed-in' flow system with all functions automatically controlled.

The butterfly valve shown in Figure 1 has a disc/stem (A) in a 316 stainless-steel one-piece design and produces bubble-tight shut-off. The valve has acetal stem bushes (B) and a double 'U' cup self-adjusting seal (C). The extended neck (D) allows for 5 mm of piping insulation. The seat is of a tongue and groove design (E) and the primary seal (F) is achieved by an interference fit. The body (G) is a two-piece wafer or lug style. The main feature of this particular type of hygienic valve is its international compatibility.

The rotor valve (Figure 2) is particularly suitable for food, beverage and pharmaceutical applications. The main benefits of this type of valve are: multi-port capability, cavity-free, high-flow, top-entry and quick-couple connection.

Figure 3 shows an air-operated remote-control changeover valve. The movement of the valve is transmitted by a piston in the actuating cylinder.

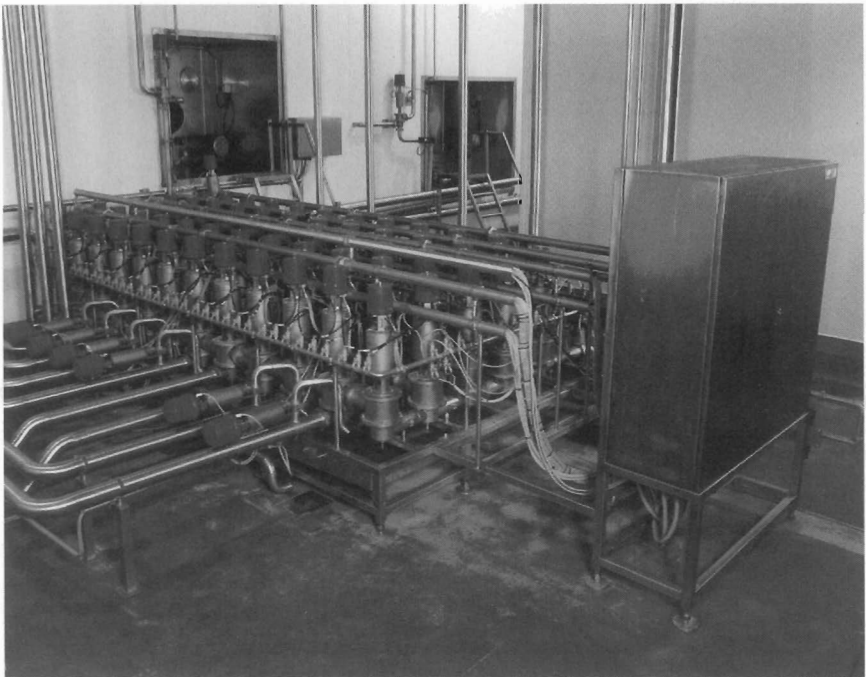
The piston is operated by compressed air arranged to open or close the valve as required, the return movement being spring-assisted to provide a fail-safe feature. The diaphragm shaft seal shown is particularly suitable for aseptic duties involving the use of steam-sterilisation procedures. The seals are manufactured from PTFE and can be used at temperatures up to 150°C (302°F).

Pharmaceuticals

The pharmaceutical industry has requirements for primary and secondary manufacturing processes. The secondary processes have standard requirements covering steam, water and quality-water supply.

The material of choice tends to be 316 stainless steel with gate- or bellows-sealed valves for steam handling, bronze-gate valves for standard water handling and diaphragm valves for quality-water handling. Primary processes require sterile valves for materials handling.

The valve shown in Figure 4 is used in the pharmaceutical drug industry and in hospitals for oxygen-supply systems. The valve is fitted with a sealed plunger mechanism to ensure there is no remnant of media in the actuating zone.



A manifold of mixproof valves installed in a dairy.

Sterile valves developed for use in bio-technology and food-processing applications are available, complete with valve actuator as shown in Figure 4. Stainless-steel body extensions at both inlet and outlet permit welding into the process line with extra-thick extensions to guarantee weld integrity.

Internal components can be steam-sterilised via permanent connections to a steam supply. Temperature-monitoring devices are attached to the valve to ensure total sterility and conventional bonnet backing is removed to avoid process contamination.



A collection of sanitary valves and pipeline components for use on process lines in food, beverages and body-care plants.

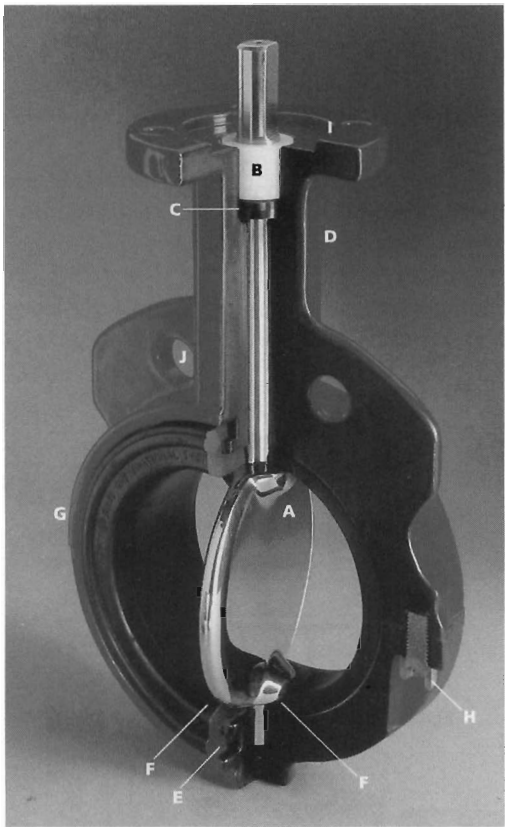


Figure 1. Butterfly valve for hygienic services.

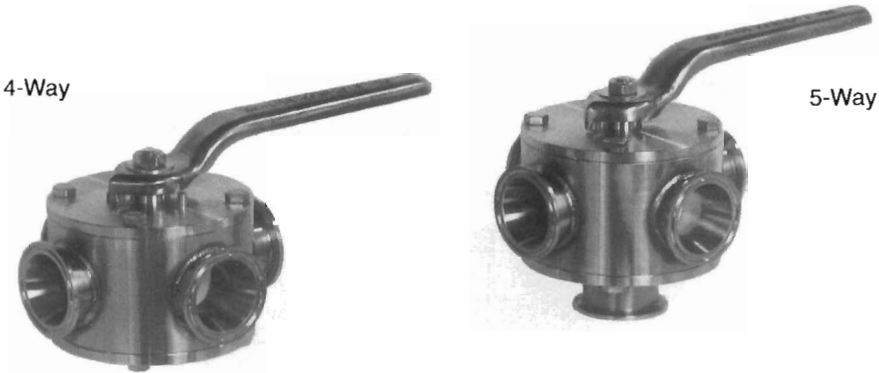


Figure 2. Multi-port rotor valves for food, beverage and pharmaceutical applications.

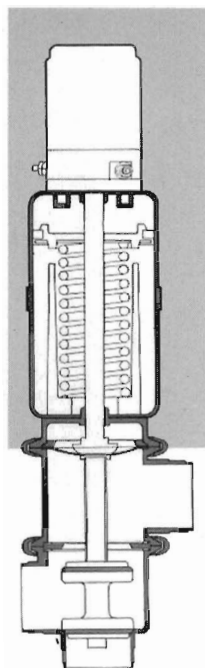


Figure 3. Air-operated hygienic-service valve.

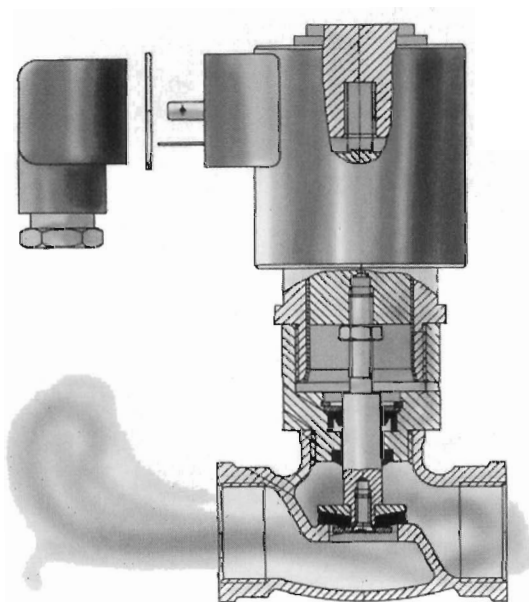


Figure 4. Pharmaceutical industry standard valve.

Stainless-steel thin-walled pipes and stainless-steel fittings are produced to various standards for use in hygienic applications. Some standards are:

BS 1864: stainless-steel milk pipes and fittings using the recessed O-ring joint (see Figure 6).

BS 3581: stainless-steel cone-joint pipe fittings.

American 3A: dimensionally similar to BS 3581 with metal-to-metal cone-type joint.

IDF (International Dairy Federation): lighter in construction than BS 1864 and employing a specially shaped rubber joint to give a flush crevice-free seal (see Figure 7). This standard is now incorporated in BS 4825 Part 4 and ISO 2853.

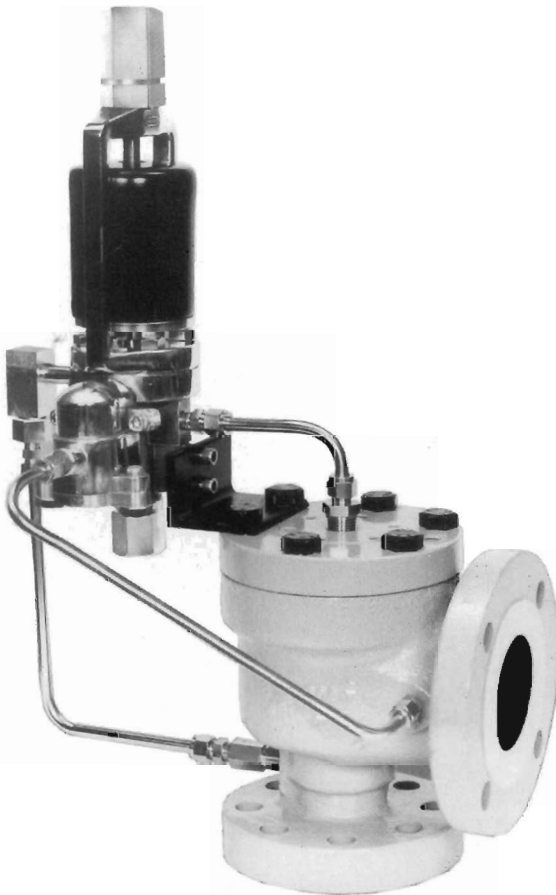


Figure 5. Sterile valve with actuator.

The most common material used for hygienic valves and pipes is 316 stainless steel or 18/10/3 stainless steel, also known as BS 316S16, with equivalent specifications as follows:

United States: AISI type 316

France: Z.8CND

Sweden: 832 SK and RRNJ44

Germany: V4A Supra

Corrosion of stainless-steel pipelines can readily occur, however, if sterilising agents based on halogens are allowed to remain in contact with the metal for extended periods. See also the chapter on Corrosion of Stainless Steel.

Glass pipelines

While plastic pipe and tube meets much of the demand for non-toxic pipelines, glass piping may be preferred, or even become essential, for sterile services. Glass is attacked by only a few reagents, which include hydrofluoric and hot concentrated phosphoric acids (both of which produce serious corrosion), superheated water and alkaline solutions.

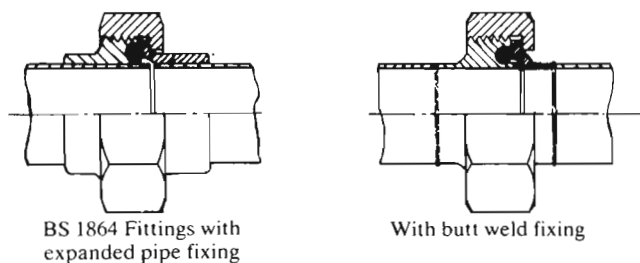
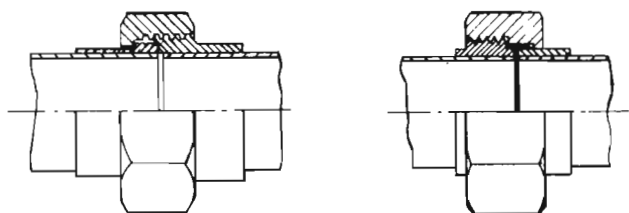


Figure 6.



BS 3581 fittings with expanded pipe fixing, also available with butt weld fixing.

Figure 7.

Cold alkaline solutions attack glasses very slowly but, as the temperature increases, the rate of attack rises rapidly. Attack also increases with increasing alkalinity. Attack by superheated water is seldom serious enough to prevent satisfactory service life from glass tubes, although the rate of attack increases with the temperature and alkalinity of the water.

Borosilicate glass pipeline systems are corrosion-resistant and neither rust nor age. They are used extensively for effluent and venting lines in accordance with DIN 1986 in scientific institutes, hospitals, and the chemical and pharmaceutical industries (Figure 8).

However, they have many other applications, e.g. for conveying numerous other liquids and gases in all branches of industry, laboratories, hospitals and in the foodstuff industry.

A typical borosilicate pipeline system has a 'slip-on' coupling which allows simple assembly without specialist knowledge and this, together with traps, laboratory drip cups and supports, constitutes a range of fittings with which virtually any installation problem can be solved.

Typical normal bores can be 40, 50, 80, 100 and 150 mm and pipe lengths between 100 and 2000 mm.

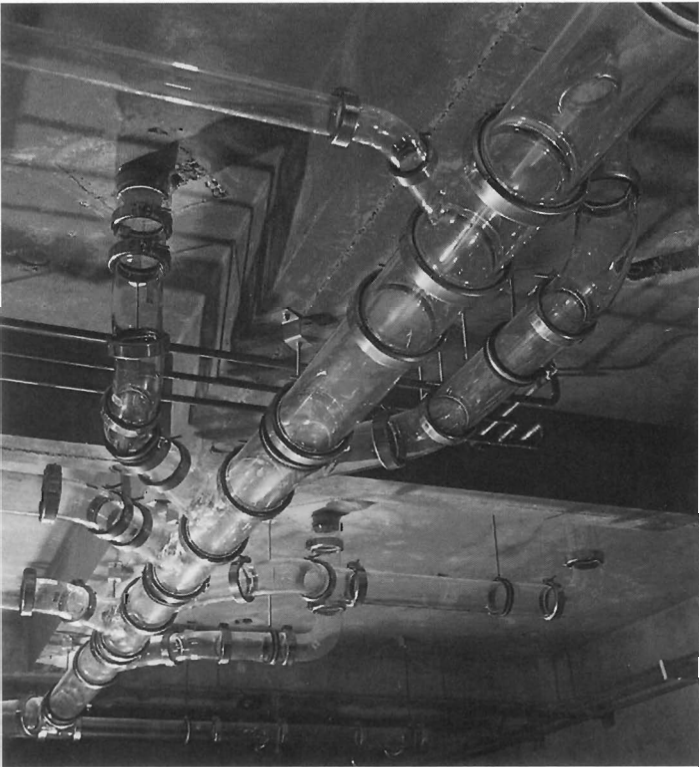


Figure 8. Borosilicate glass piping system.

The glass components are connected together by means of a coupling consisting of a single-bolt stainless-steel clamp which produces a tight connection. Within this clamp is a flexible nitrile rubber insert, which positively grips the bead flanges of the glass parts to be connected.

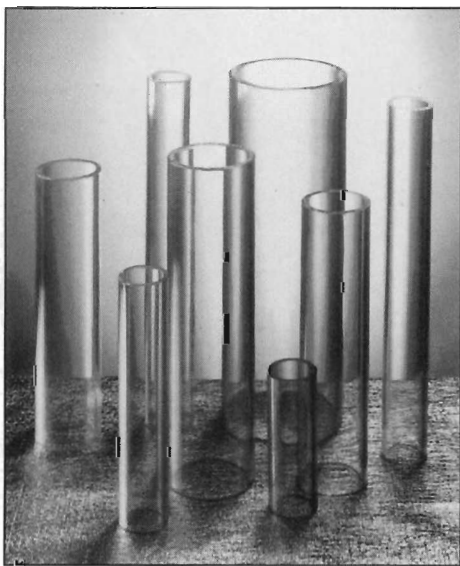
At the actual sealing point, there is a PTFE insert which has the same chemical resistance as the glass itself, thus the medium only comes into contact with borosilicate glass and PTFE.

The most complicated pipeline installation can be assembled and the low weight of glass (density only about one-third that of cast iron) is an advantage which aids installation under ceilings and other inaccessible areas. Glass piping is fragile and does require safeguarding, particularly if used in pressurised applications.

Maintenance costs can be reduced by visual inspection through the pipe, which enables rapid location of potential build-up and blockages. The use of rods to remove blockages is acceptable provided that non-metallic ferrules and attachments, e.g. nylon, are used.

See also the chapter on Thermoplastic Pipes.

Thermoplastic pipes have their attractions because of their chemical inertness and a structure which does not harbour bacteria. Not all such materials are hygienic in the sense that they are free from tainting the product, even PTFE not being ideal for handling foodstuffs. All such materials, too, suffer from relatively low maximum-service temperatures, which can make certain types unsuitable for sterilisation via cleaning in place. Certain



Rigid clear PVC pipe.

food products are, however, successfully handled by elastomeric pipes (hoses) or rubber-lined pipes.

Plastic-lined pipe is a frequent choice in food and beverage processing and pharmaceutical facilities. Entire plastic systems that meet FDA requirements can be assembled using plastic-lined pipe and valves.

Polyvinylidene chloride (PVDC) is suitable as a pipe-lining material for high purity needs to 80°C (175°F), where stream-purity protection is critical.

Another successful plastic piping material for pharmaceutical and biotechnology industries is unpigmented natural polyvinylidene fluoride (PVDF). PVDF plastic piping is one of the most chemically inert engineering polymers. It is an excellent material for all types of ozonation systems, for example, particularly where both purity and strength are required.

Clear Schedule 40 rigid PVC is a good choice for sight-glass and dual-contaminant applications. Made from non-toxic materials conforming to FDA standards, it has found success in wet- and dry-food processing, bakery products, medical/hospital uses and in the cosmetics industry. It is a cost-effective alternative to copper, stainless steel and glass piping. It is not recommended for compressed-gas applications or pipe threading.

Steam Services

Steam is water in the vapour phase and is one of the oldest industrial tools.

The first requirement in steam production is to add heat to water until it reaches its boiling point. It is then necessary to add a much greater quantity of heat to convert the water to steam.

Steam allows the energy of fuel burned in the heat source or boiler to be carried to some other point where it can either provide mechanical energy through an engine or provide heating.

In all types and sizes of oil and chemical plants, energy is used for process heating, power generation and driving pumps and compressors.

In many refineries, primary steam is obtained by burning waste products in the boilers. Although steam is the traditional means of conveying heat, there are a number of alternatives including:

- high- and medium-pressure hot water.
- high-temperature oils.
- electric heating.
- mechanical agitation.

It is not the remit of this handbook to discuss the merits of steam or alternatives, neither is it appropriate to discuss the subject of steam in its full capacity.

This subject is well documented in another more technical publication and in some specific manufacturers' literature and publications.

Steam distribution

The most important link between a central steam source and the steam user is the steam-distribution system. A typical steam circuit is shown in Figure 1.

The steam flow in a circuit is caused by condensation of steam which produces a pressure drop. This induces the flow of steam through the pipes to where the heat energy is required.

In operation when the steam outlet (crown) valve is opened, steam passes immediately from the boiler into and along the main pipes. The pipework is cold initially so heat is transferred to it by the steam. The air surrounding the

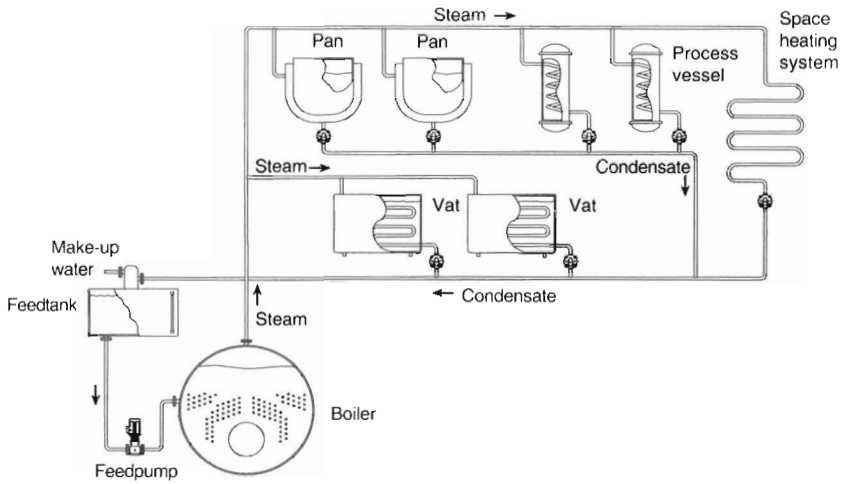


Figure 1. A typical steam circuit.

pipes is cooler than the steam, so the pipework will begin to lose heat to the air. This causes the steam immediately to condense and fall to the bottom of the pipe. It is then carried along with the steam flow and by gravity owing to the gradient in the steam main which normally falls in the direction of steam flow. The condensate is drained from the lowest points in the pipeline.

By continuously feeding more fuel and water into the boiler, a continuous flow of steam is maintained to make up for the water which has already evaporated into steam.

The condensate is usually returned to the boiler feed tank.

The pressure at which the steam is to be distributed is to some degree determined by the point of usage on the plant needing the highest pressure. Steam at a higher pressure occupies less volume per kilogram than steam at a lower pressure.

Steam boilers

Boilers are the most important part of the steam circuit. A boiler is a vessel in which the heat energy from a fuel is transferred to a liquid. In the case of saturated steam, a boiler also provides heat energy to produce a phase change from liquid to vapour. Steam boilers come in all sizes to suit both large and small applications and operate using different fuels, including commercial waste, oil, gas and coal.

The choice of fuel is largely dependent on the tariff given to each type of fuel. Boilers can operate on just one or on two types of fuel (e.g. oil and gas). A typical package boiler is shown in Figure 2.

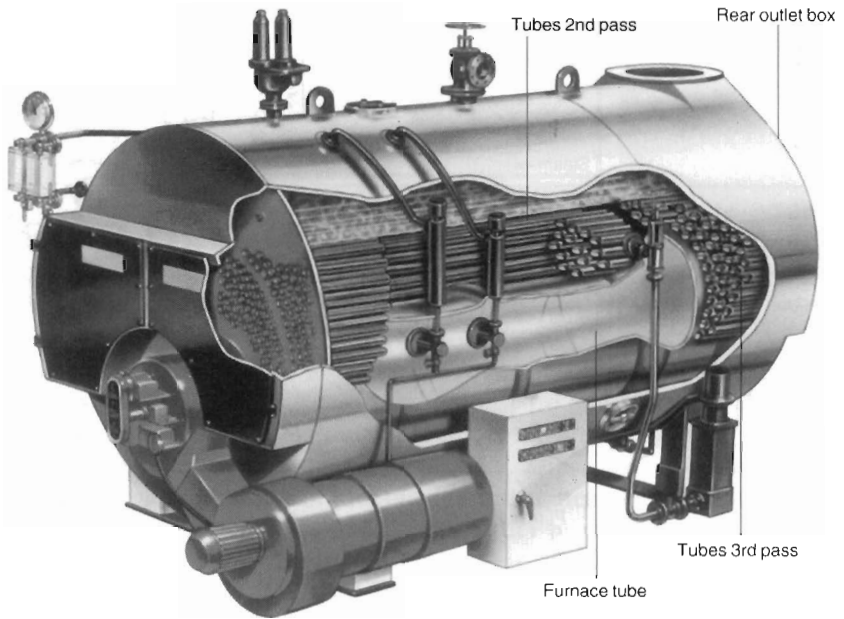


Figure 2. A typical package boiler.

Superheated steam

Steam produced from the outlet of a shell-type boiler or from the steam drum of a water-tube boiler can only be saturated steam. Water-tube boilers are often required to produce superheated steam by passing saturated steam from the steam drum through another set of tubes inside the main furnace area, where it is heated up beyond its saturation temperature to a gas (superheated steam). Where superheated steam is required, a boiler incorporating superheating tubes is essential.

Safety valves

An important boiler fitting is the safety valve. Its function is to protect the boiler shell from over-pressure and subsequent explosion.

There are many types of safety valves fitted to steam boiler plant but they must all meet the following criteria:

- The minimum bore of a safety valve connected to a boiler must be 20 mm.
- The total discharge capacity of the safety valve(s) must be at least equal to the drum and at 100% capacity of the boiler.

- The full rated discharge capacity of the safety valve(s) must be achieved within 110% of the boiler design pressure.
- The maximum set pressure of the safety valve(s) shall be the design (or maximum permissible working pressure) of the boiler.
- There must be an adequate margin between the normal operating pressure of the boiler and the set pressure of the safety valve.

A typical boiler safety valve is shown in Figure 3.

Double safety valves are commonly found on boilers with an evaporative capacity of more than 3700 kg/h.

Stop valves

A stop valve (crown valve) must be fitted to a boiler in order to isolate the steam boiler and its pressure from the process or plant.

Typically, stop valves used are generally angle-pattern globe valves of the screw-down type. Cast-iron valves should not be used for this application.

The stop valve is not designed as a throttling valve and should be fully open or closed. It should always be opened slowly to prevent any sudden rise in downstream pressure and associated water hammer. The valve should be of the 'rising handwheel' type in order that the valve position can be easily seen. An inductor fitted to the valve also assists this procedure.

Isolating valves, usually screw-down globe valves with disc-check valves sandwiched between the flanges of the two stop valves, are used on multi-boiler applications.

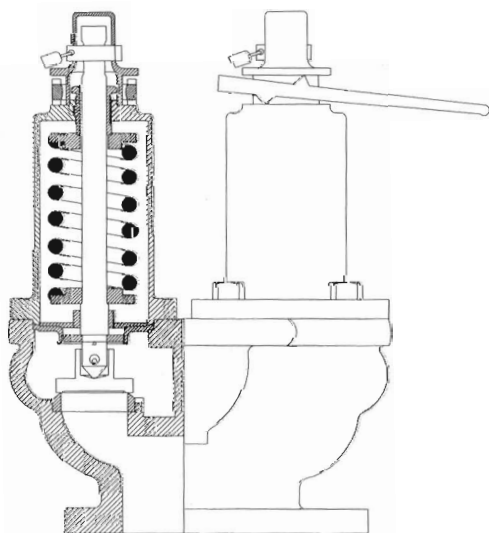


Figure 3. A typical boiler safety valve.

Feed-check valves

These are installed in the boiler feed-water line between the feed pump and boiler. A boiler feed stop valve is fitted at the boiler shell.

Bottom blow-down valves

Steam boilers should be fitted with at least one bottom blow-down valve at a point as close as possible to where sludge or sediment is likely to accumulate. Blow-down valves should be key operated or automatically controlled by timers and electronic interlocks.

Air vents and vacuum breakers

Simple cocks and pressure-balanced air vents are designed to purge air from the steam space (Dalton's Law). Vacuum breakers are fitted on the boiler shell. They are used when a boiler is taken off-line and the steam space condenses and leaves a vacuum that can result in damage to boiler flat plates and leaks from inspection doors.

System valves

In addition to both safety and control valves, butterfly valves are used in steam-pipeline systems when tight shut-off is required.

Check valves are used for the protection of reverse flow in pipelines. They are typically of the wafer pattern up to 40 bar.

Ball valves to over 60 bar are commonly installed throughout the system.

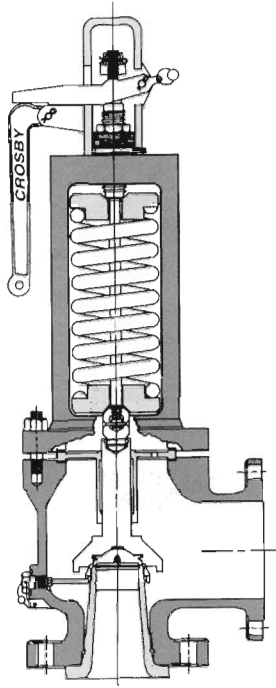
Bellows-sealed stop valves are ideal for high-pressure and high-temperature applications.

Pressure reduction

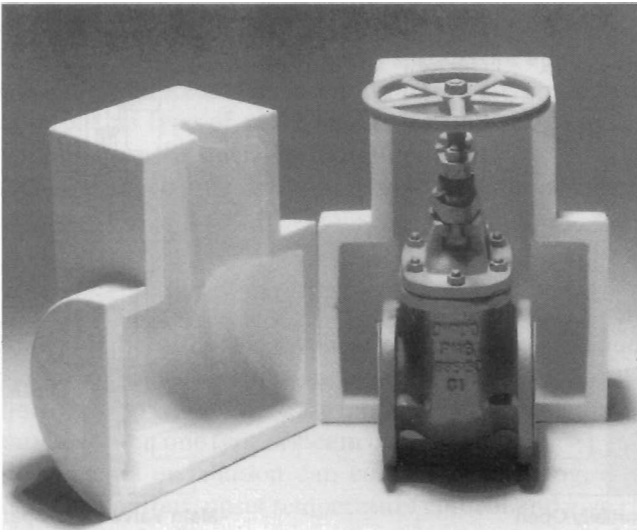
Steam generators produce steam at a pressure, temperature and volume not generally acceptable to a consumer. As a consequence, it is necessary to apply steam conversion (i.e. change one, two or all three parameters) in the steam-distribution systems of power plants for public safety, heating power plants, industrial power plants and in the chemical and process industries.

Conventional practice is to reduce steam pressure in a reducing valve station (see Figure 4), and reduce temperature further down the line with desuperheaters operating on a variety of principles, e.g.:

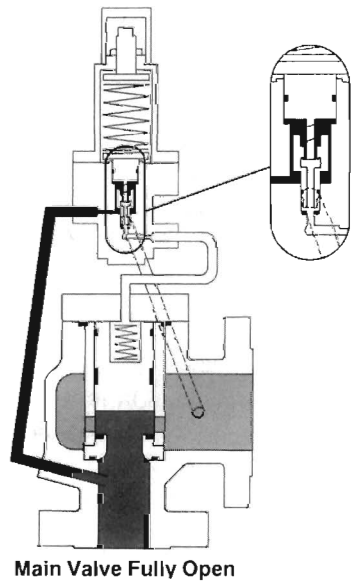
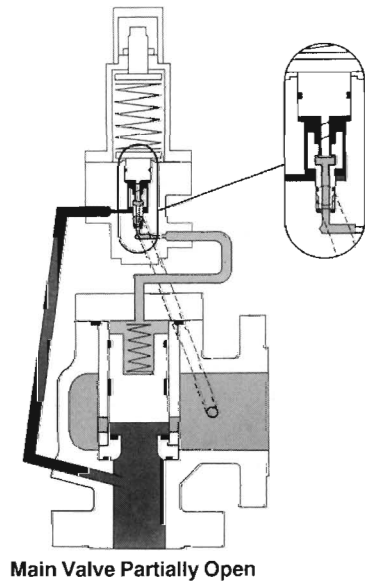
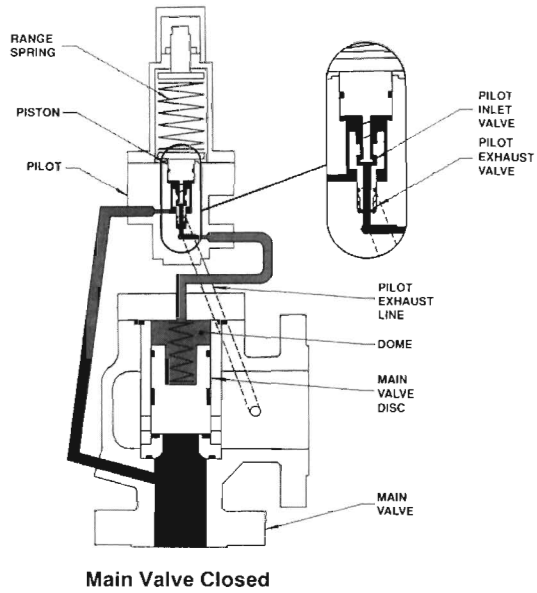
- (i) direct water injection into the low-pressure steam line
- (ii) applying saturated steam to the hot steam
- (iii) steam cooling by a separate steam cooler.



High-performance open-bonnet safety valve for saturated and superheated steam service.



Rigid moulded insulated valve cover.



Non-flowing modulating pilot-operated pressure-relief valve.

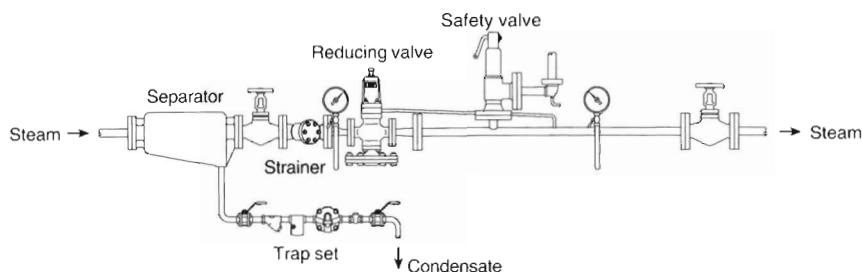


Figure 4. A typical pressure-reducing valve station.

The two processes of pressure reduction and desuperheating can also be accomplished either simultaneously or sequentially in a steam converter valve.

Details of steam-control valves and steam-relief valves are shown in Tables 1 and 2.

Valve operation (Figure 3)

Steam-conditioning valves

Steam-conditioning valves are primarily designed for pressure and temperature control of steam. The first valve was developed in 1929 and had butterfly pressure control.

Steam-conditioning valves (Figure 5) combine pressure and temperature reduction in a single body and are particularly suitable for applications such as: turbine bypass valves, process steam-conditioning valves and pressure-relief valves. The use of this type of valve means less rigorous requirements for piping downstream from the steam-conditioning valves.

In situations where cooling without pressure reduction is required, a desuperheater is used.

Steam traps

In any steam installation, the effectiveness of the steam-distribution system and the steam-using plant depends to a large extent on the correct selection and application of steam traps. There is a tendency to underestimate this point in order to standardise on one type of steam trap.

Wrong selection and installation can cause waterlogging, damage plant performance, reduce output, upset temperature control and give rise to water hammer. Wrong sizing is often the root cause of pressurised condensate lines, short trap life and high maintenance costs.

Table 1. Steam control valves: applications

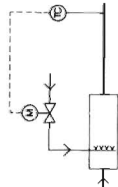
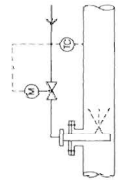
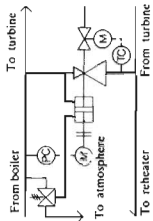
Temperature controlling equipment		Combined steam converting and safety valve	
Superheated steam cooler	Nozzle injectors		
Circuit 	Circuit 	Circuit 	
Application: For special temperature-control tasks in industrial power plants.	Application: For special temperature-control tasks in industrial power plants.	Application: For high-pressure (HP) bypass stations during start-up and bypass operation, primarily in public utility power plants.	
Typical design: With drawn pipes, welded flanged (to DIN, ANSI, etc.).	Typical design: Nozzle arrangement as required. Division of the spray water among several immersion pipes possible depending upon purpose.	Typical design: Forged, welded connection (to DN, ANSI, etc.). Type 500: inlet at side. Type 600: inlet from below.	
Remarks: Low pressure loss: no additional atomising steam. Low noise level. No moving parts: simple maintenance. Fitted in all positions. Sizes: DN 125 to DN 1600 (mm).	Remarks: High-duty nozzles for easy installation in every pipeline. No moving parts, simple maintenance. DN > 50 (mm).	Remarks: In conjunction with the auxiliary control system, it functions as a safety valve. No external control media are used, only the existing live steam. With an electric pressure-control system, the valve functions as a normal control valve.	

Table 1. Continued

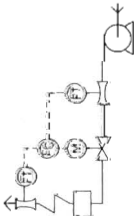
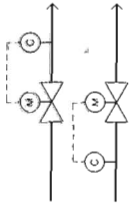
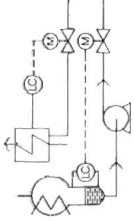
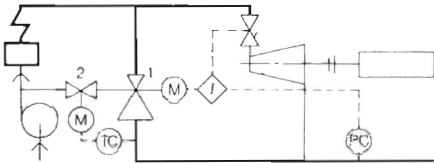
Feed-water control valves	Conventional control valves	Level control valves
Circuit	Circuit	Circuit
		
Application: For controlling all or part of the feedwater; also available in a special version as a boiler filling valve.	Application: In plants for pressure control of oil, water, steam, etc. As auxiliary control valve, drain valve of start-up flash tank.	Application: On high- and low-pressure (HP and LP) feed-water tanks, condensers, etc.
Typical design: Cast, forged, straight-through valves, angle valves, welded connections, flanges (to DIN, ANSI, etc.).	Typical design: Cast, forged, welded pipe construction, straight-through valves, welded connections, flanges (to DIN, ANSI, etc.).	Typical design: Cast, forged, straight-through valves, angle valves, welded connections, flanges (to DIN, ANSI, etc.) Valve parameters (DN, PN) as required.
Remarks: Versions available for all capacities encountered in practice. DN 50 to DN 500 (mm).	Remarks: Typical sizes and pressure ranges, From DN 15 to DN 1500 (mm). From PN 10 to PN640 (bar) (145 to 9300 lbf/in ²).	Remarks: For plant-specific problems with evaporating media (cavitation, etc.).

Table 2. Steam relief valves: applications

Steam converting station	
Steam converting valve	Spray water valve
 1 Steam converting valve 2 Spray water valve	
Application: In all thermal systems where steam pressures and temperatures are to be simultaneously reduced, e.g. in steam-distribution systems in industrial plants and, in particular, as a turbine bypass in power plants.	Application: For steam converting plants, combined steam converting and safety stations and temperature-control equipment.
Typical design: Cast, forged, straight-through valves, angle valves, welded connections, flanges (to DIN, ANSI, etc.).	Typical design: Cast, forged, straight-through valves, angle valves, welded connections, flanges (to DIN, ANSI, etc.). with single- or multi-stage perforated cage trim depending upon the pressure drop to be handled.
Remarks: Saving of space and improved control quality as a result of simultaneous pressure reduction and desuperheating in one single valve. Maximum design data for valves supplied to date: 700 t/h, 276 bar (4000 lbf/in²) 565°C (1050°F).	Remarks: DN 15 to DN 150 (mm) and higher.

There are various types of steam trap (Figure 6), which include:

- **Thermodynamic traps:** suitable for mains drainage and for draining tracers or jacketed lines. Its simple design with one moving part, a flat disc, makes it a popular choice. This type of trap can withstand severe water hammer and freezing.
- **Bimetallic traps** operate by the movement of a bimetallic strip with temperature change. These traps work over a wide range of pressures but with a degree of waterlogging.
- **Balanced-pressure traps** have capsules manufactured in stainless steel for corrosion resistance. They also have a high resistance to water hammer and are not affected by back-pressure. The balanced-pressure principle is now widely accepted for applications where thermostatic steam traps can utilise sensible heat in the condensate and reduce flash steam losses.

- Fixed-temperature discharge traps: one of the balanced-pressure range of steam traps for discharging condensate at just below steam temperature when installed in one altitude, or at a fixed temperature when installed in reverse. A condensate degree of sub-cooling and perhaps waterlogging is implicit in the use of a trap of this pattern.
- Mechanical steam traps: probably the most popular choice of steam trap is the ball-float type.

First launched in the 1940s, the ball-float steam trap will pass condensate as soon as it reaches the trap but will hold back steam and

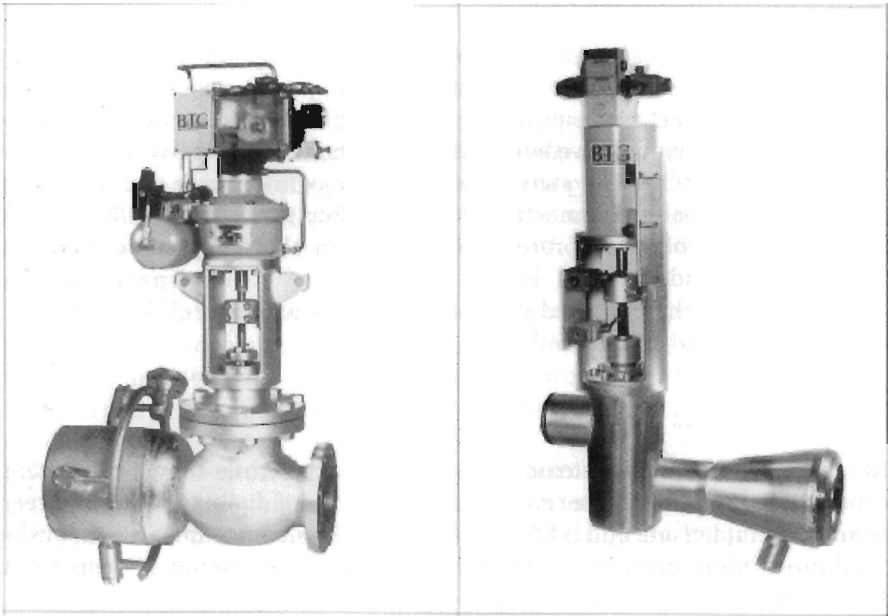
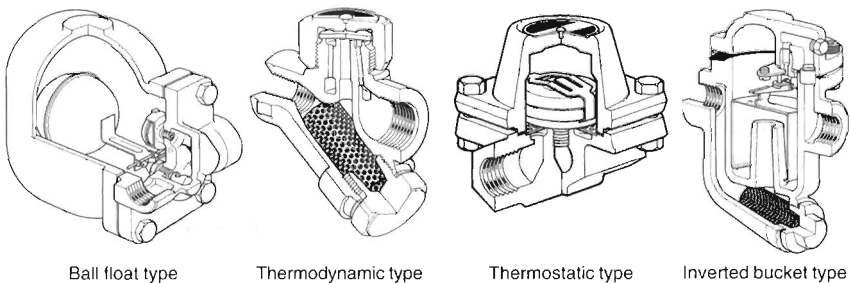


Figure 5. Steam conditioning valves.



Ball float type

Thermodynamic type

Thermostatic type

Inverted bucket type

Figure 6. Steam traps.

adjust itself automatically to a change in load. It is particularly suitable for draining plant which must be kept clear of condensate at all times. Automatic air vents and steam-lock release valves may also form an integral part of this type of steam trap. Ball-float steam traps generally have a high discharge capacity, whether installed in a horizontal or vertical position.

Inverted-bucket steam traps normally give an intermittent discharge and are more suited to superheat conditions because they are sensitive to density. If bucket traps are used on plant which will frequently be called upon to start from cold, separate air vents should be provided in by-passes. Bi-metallic or balanced-pressure elements are more suited to this requirement.

All mechanical steam traps are comparatively bulky and, since they contain water, they must be protected from freezing in exposed conditions.

- **Swivel-connector steam traps:** this type of trap is designed for ease of fitting and removal without the need to break the pipework or disrupt plant operation. The swivel quick-release connector is a permanently installed pipeline connector with a leak-free joint. Three different trap options can be incorporated into this system: thermodynamic, balanced pressure and inverted bucket. This type of steam trap is ideal in situations where altered plant or process conditions call for a different type of trap to be installed.

Steam-trap monitors

This type of unit enables steam traps to be checked while they are working. Typically, it consists of a sensor chamber capable of distinguishing between steam and condensate and is fitted upstream of the steam trap. It is suitable for continuous monitoring and can operate on saturated steam systems up to 32 bar. The unit can be used with any type of steam trap.

Air vents

The three primary barriers to heat transfer are films of water, air and scale. By far the most resistant to heat transfer is air. In fact, air is more than 1500 times more resistant to heat transfer than iron or steel and no less than 13,000 times more resistant than copper.

Thermostatic air vents automatically open to air and gases, but shut against steam. They discharge air full bore on start-up and open during running whenever air collects, irrespective of steam pressure. Air vents should be located furthest away from the steam inlet because this is where air tends to collect. Where possible they should be fitted at all high points in the system. Manually-operated air cocks are not suitable when dealing with air and uncondensable gases that are mixed in with steam.

Balanced-pressure air vents are the most widely accepted type of air vent, because they operate close to the steam-saturation temperature and can therefore differentiate between pure steam and air/steam mixtures. They have a high resistance to superheat and water hammer.

Pipeline sizing

Pipe sizes may be chosen on the basis of either

- fluid velocity
- pressure drop

There is often a tendency when determining pipe sizes to be guided by the size of connections on equipment to which they will be connected. The desired volumetric flowrate may not be achieved if the pipework is sized in this way.

If the pipe is too small, then high pressure drop and steam starvation at the using end will result. If the pipe is too large, the pipes will be more expensive than necessary, a greater volume of condensate will be formed due to greater heat loss and increased running costs will result. The quality of the steam will also be poorer.

The most common pipe standards used when sizing pipe are those derived from API. In Europe pipe is manufactured to DIN standards. Other terms used include Blue band and Red band referred to from BS 1387 regarding pipe thickness. Red is commonly used for steam-pipe applications and Blue for air-distribution systems.

If pipework is sized on the basis of velocity, then calculations are based on the volume of steam being carried in relation to the cross-sectional area of the pipe.

Pipe is sized on the 'pressure drop' method by using the known pressure at the supply end of the pipe and the required pressure at the point of use.

Boiler-feed calculations and steam-flow calculations are more comprehensively covered in a preceding section of this handbook. In addition, a number of specialist steam equipment manufacturers have produced some excellent publications that cover this subject in considerable detail.

In any steam main, some of the steam will be condensed due to radiation losses and this water must be drained out otherwise corrosion and water hammer will result. In addition, the steam will become wet as it picks up water droplets.

Steam-pipe mains should be run with a fall of not less than 40 mm in 10 m ($1\frac{1}{2}$ in in 10 ft) in the direction of the steam flow, to ensure that both steam and condensate run in the same direction. Drain points are installed in the line to collect and remove water.

Steam-distribution systems will often give more trouble than any other piped service because of the failure to recognise that the pipework contains not only steam but water and air. Saturated-steam lines should be drained at

regular intervals, at all low points where the condensate can collect. Drain points at intervals of 30–50 m (100–150 ft) are usual and they are most effective where pipework changes direction.

Pipework should be arranged so that pockets where water can collect are avoided. Globe valves of under-and-over construction can also form a weir and prevent condensate from flowing to the next drain point. If the valve is fitted on its side, this can usually be avoided. It is important also to note that branch lines are normally much shorter in length than the steam mains. Sizing branch lines on the basis of a given pressure drop is less convenient on short lengths of pipe.

Branch-line pipe sizes are normally selected from a table based on pipeline capacities at specific velocities.

Branch connections should always be taken from the top of the main so that the driest steam is taken.

Steam tracing

The temperature of process liquids being transferred through pipelines must often be maintained to meet the requirements of the process, to prevent thickening and solidification or simply as an anti-frost measure. This is achieved by the use of jacketed pipes or by attaching to the product line one or more separate tracer lines, each carrying a heating medium such as steam or hot water.

External tracer lines are simple and cheap to install. The simplest form of tracer is one that is clipped or wired onto the main product line. Maximum heat flow is achieved when the tracer is in tight contact with the product line. The material for the tracer line should always be chosen to avoid electrolytic corrosion at any contact points.

Tracer pipes can generally be wired on but it is better to use either galvanised or stainless-steel bands. A practical method is to use a packing-case banding machine. Where temperature difference between the tracer and the product is low, the tracer may be welded to the product line using either short-run or continuous welds for maximum heat transfer.

For maximum heat transfer it can be an advantage to use a heat-conducting paste. The paste can be used to improve heat transfer with any of the clipping methods described. The surfaces should be wirebrushed clean before the paste is applied.

Spacer tracing involves the use of insulating material between the tracer and the product pipe to avoid local hot spots on the pipe if the product being carried in the line is sensitive to temperature.

If pipes are to be insulated, the insulation should cover both product line and tracer but it is important that the air space remains clear. The insulation should be properly finished with a waterproof covering. Most of the sizing of external tracers is done by rule of thumb but the problem is—what rule? and whose thumb? There are widely differing opinions on lay-out as well.

Fire-Safe Valves

Defining industrial fires is a difficult task.

They can range from smoking, oxygen-poor, low-temperature fires with their resultant low heat flux to the extreme of a hydrogen-jet fire with flame temperatures exceeding 2800°C (5000°F).

No valve is an entity by itself in a fire. The entire system has to be considered and includes effects on pipe supports, pressure-retaining bolting tanks, and concrete structures.

The API (American Petroleum Institute) fire tests were designed specifically for valves in oil and gas production plants. Over many years the tests have been refined and adapted for many different industrial plants. API 607 (Rev 4), API 6FA and BS 6755, BS 5146 are the accepted standards for fire tests today and adopt the same procedures. Other test standards and procedures include OCMA FSV-1, Exxon BP3-14-1, FM 6033 and API RP6F. Many users have also established their own corporate standards for fire-safe valves.

A strategy of fire fighting that appears to be universal as far as fire duration is concerned is that if the fire is not beaten in one half hour, a withdrawal and containment policy is instituted. This comes from structural component failures such as pipe-rack collapse, flange-bolt failures and concrete eruptions due to water of hydration changing to steam.

Based on the limiting factors of ancillary equipment, a test duration of one half hour was established.

The basis of all fire tests is that a pressurised valve must operate after being burned at a specified high temperature for a specified period and leakage after burning (which will destroy soft seals) must remain within specified limits.

Valve-open or valve-closed test

Soft-seated valves which employ seats on both the upstream and downstream side of the obturator will trap fluid in the cavity formed by these seats and the pressure shell. If this fluid is an incompressible liquid, increases in temperature will cause the pressure of the trapped fluid to rise dramatically.

Calculation and rather simple experiments both show that a trapped hydrocarbon liquid will increase in pressure by 12 bar for every °C of temperature rise.

This is a known problem that has been so overlooked that paragraph 2.3.3 of the American National Standards B16.4, 'Valves—Flanged, Threaded and Welding end', warns about the effects of thermal expansion of fluids trapped in double-seated valves.

While some ball-valve designs are capable of automatically relieving this cavity pressure when in the closed position to the upstream side, others may not.

In OCMA FSV-1, the test did not evaluate the valve's resistance to cavity-pressure rise as conditions with the valve open and a vent hole in the ball stem slot did not represent those of a closed valve. API insists on a cavity-filled closed test.

Specific API test requirements are:

- (a) The valve shall be tested in the closed position with water, with the stem and bore in the horizontal position. Check valves will be tested in their normal operating position.
- (b) The valve will be uniformly enveloped in flame having a temperature of 761–871°C (1400–1600°F) average of two thermocouples, one located 25 mm (1 in) below the valve and the other 25 mm (1 in) from the upper stem packing box on the horizontal centreline. No reading shall be below 704°C (1300°F). Piping upstream of the test valve larger than 25 mm (1 in) nominal pipe size or one half of valve nominal pipe size (whichever is smaller) must be enveloped in flame for a distance of at least 152 mm (6 in).
- (c) The end connection piping-to-valve joint leakage (flanged, threaded or welded) is not considered a part of this test and is not included in the allowable external leakage. For the test, it may be necessary to modify this joint to eliminate leakage.

Suggested systems for fire testing to API specifications are shown in Figures 1 and 2. Figure 2 is a schematic outline for systems using compressed gas as the pressure source. Test procedure is as follows:

- (i) Open valve(s) (items 5 and 6) at water source, and any necessary vent valves (item 17) to flood the system and purge the air. The test valve may have to be placed in the partially-open position in order to completely flood the valve body.
- (ii) Close fill valve (item 5) and vent valves (item 17), and close the test valve (item 11). The system upstream of the test valve should be completely water-filled and the system downstream shall be drained.
- (iii) Pressurise the system to the appropriate pressure from Table 1. Maintain this pressure during all testing. Record the reading on the calibrated sight gauge (item 4). Empty the graduated downstream container (item 19).

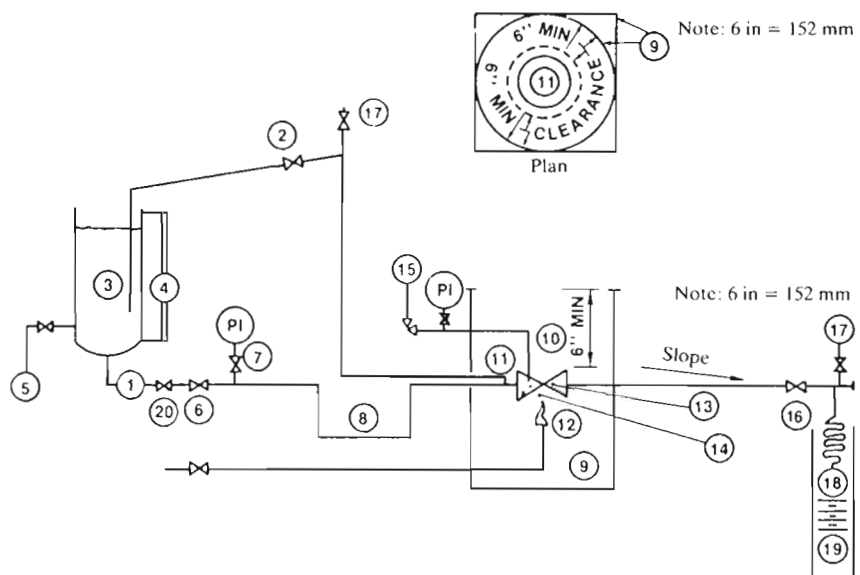
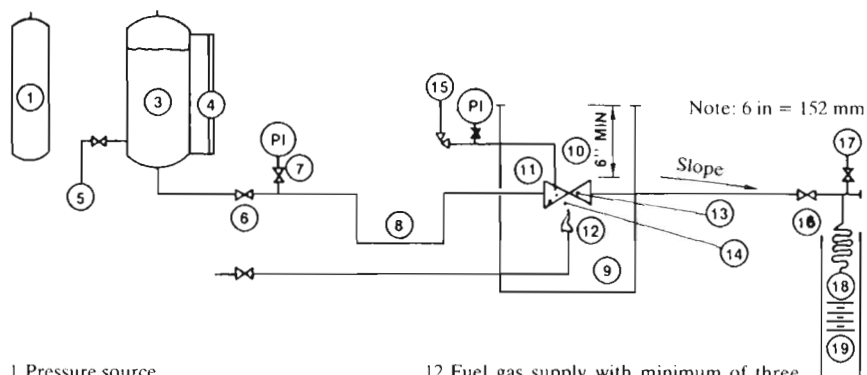


Figure 1. System using a pump as the pressure source.



1. Pressure source.
2. Pressure regulator and relief.
3. Vessel for water.
4. Calibrated sight gauge.
5. Water supply.
6. Shutoff valve.
7. Pressure gauge.
8. Piping arranged to provide vapour trap.
9. Enclosure for test – horizontal clearance between any part of the valve and the enclosure shall be a minimum of 152 mm (6 in).
10. Minimum height of enclosure shall be 152 mm (6 in) above the top of the valve.
11. Test valve mounted horizontally with stem in horizontal position.

12. Fuel gas supply with minimum of three burners located at 120°
13. Flame temperature thermocouple – located 25 mm (1 in) from upper stem packing box on horizontal centreline.
14. Flame temperature thermocouple – to be located 25 mm (1 in) below the centre of the valve body.
15. Pressure gauge and relief valve (if required) connected to centre cavity of valve.
16. Shutoff valve.
17. Vent valve.
18. Condenser.
19. Calibrated container.
20. Check valve.

Figure 2. System using compressed gas as the pressure source.

- (iv) Open fuel supply, establish a fire, monitor the flame temperature, and when the average of the two thermocouples (items 13 and 14) reaches 761°C (1400°F) start the test. Maintain the average temperature between 761 and 871°C (1400 and 1600°F) for the test duration. No reading shall be less than 704°C (1300°F).
- (v) Record instrument readings (items 7, 13, 14 and 15) every 2 min for the test duration.
- (vi) At the end of the test duration (15 or 30 min), shut off the fuel.
- (vii) Immediately determine the amount of water collected in calibrated container (item 19) to establish total through-valve seat leakage. Continue recording the amount of water collected for use in establishing the external leakage rate. If the test valve is of the upstream sealing type, the volume of water that is trapped between the upstream seat seal and the downstream seat seal (when the valve is closed) shall be determined before the test is started and identified in the test report. It is assumed that during the test this volume of water would move through the valve, past the downstream seat seal and be collected in the calibrated container. This volume has not actually leaked past the upstream seat seal, so it may be deducted from the total volume measured in the downstream calibrated container when determining the through-valve leakage.
- (viii) Allow the test valve to cool to 93°F (200°F) or less. Use temperature-sensitive crayons or other suitable means to indicate valve-body temperature near the thermocouples (items 13 and 14). Record the level in the sight gauge (item 4). Use the initial and final readings to determine total leakage during the test.
- (ix) Close the shut-off valve (item 16) and operate the test valve against test-pressure differential (Table 2) to the full-open position.
- (x) Measure and record external leakage for a minimum of 5 min after valve is in the full-open position at test pressure. Divide the total external leakage by the duration of the test in minutes to obtain the external

Table 1. Summary from SGS Yarsley fire-test report

Leakage rate (ml/min)	Through leakage rate	Maximum allowed	External leakage rate	Maximum allowed
Burn period	Zero	5600	Zero	1400
Cool down				
Low hydrostatic pressure test	Zero	560	Zero	280
High hydrostatic pressure test			Zero	2800

leakage rate. The test system, excluding the test valve, may be adjusted during the test period to keep the test within the limits specified herein.

Fire-safe ball valves

Fire-safe ball valves are manufactured for applications in explosive and fire-risk environments, and are specifically designed to prevent the spread of fire. The fire-safe ball valve shown in Figure 3 is of the floating ball type fire-tested to API 607, API 6FA and BS6755 Part 2. This type of valve is suitable for use in the oil, chemical, petrochemical and pharmaceutical process industries.

The floating ball design relies on the downstream movement of the ball due to pressure differential to effect a seal against a resilient seat ring. The valve employs a double-stage sealing arrangement and independently loaded graphite fire-safe packing that remains unaffected by any deterioration of the main PTFE chevron packing set under fire conditions.

Butterfly valves

Pyrogenic and fire-safe butterfly valves are usually certified fire-safe to BS6755 Part 2 and API 6FA. They have inherently fire-safe primary metal/metal sealing and bubble-tight shut-off in both directions.

Typical fire-test results for this type of valve are shown in Table 1, which refers to the triple-offset metal-seated valve shown in Figure 4. It has zero-leakage

Table 2. Test pressure during fire test

Type of valve	Valve rating (PN) ^a	Test pressure	
		lbf/in ²	bar
Spec 6D valves	150 (20)	210±10%	(14.5±10%)
	300 (50)	540±10%	(37.2±10%)
	400 (64)	720±10%	(49.6±10%)
	600 (100)	1080±10%	(74.5±10%)
	900 (150)	1620±10%	(111.7±10%)
	1500 (250)	2700±10%	(186.2±10%)
	2500 (420)	4500±10%	(310.3±10%)
	(bar)		
Spec 6A valves	2000 (130)	1500±10%	(103.4±10%)
	3000 (207)	2250±10%	(155.1±10%)
	5000 (345)	3750±10%	(258.6±10%)
	10,000 (690)	7500±10%	(517.1±10%)
	15,000 (1034)	11,250±10%	(775.7±10%)
	20,000 (1379)	15,000±10%	(1034.2±10%)

^a(PN) is the pressure class designation utilised in ISO (International Standards Organisation) documents.

performance, fire-tested to API 6FA and BS6755 Part 2. The block and bleed seal configuration consists of two metal-laminate seals with an intermediate bleed channel, which connects to the bleed purge port in the valve body when the disc/segment is in the closed position. The valve is designed as a replacement for gate, globe, ball or plug valves.

Some advantages claimed of a triple-offset segment valve over ball, gate and plug valves are given in Table 3.

Flame arresters

Flame arresters are employed as safety devices to extinguish flames in pipelines, ducts and vents carrying flammable gases or vapours. They prevent fire spreading to other parts of the system, helping to avoid extensive and costly damage to plant and equipment and the risk of personal injury.

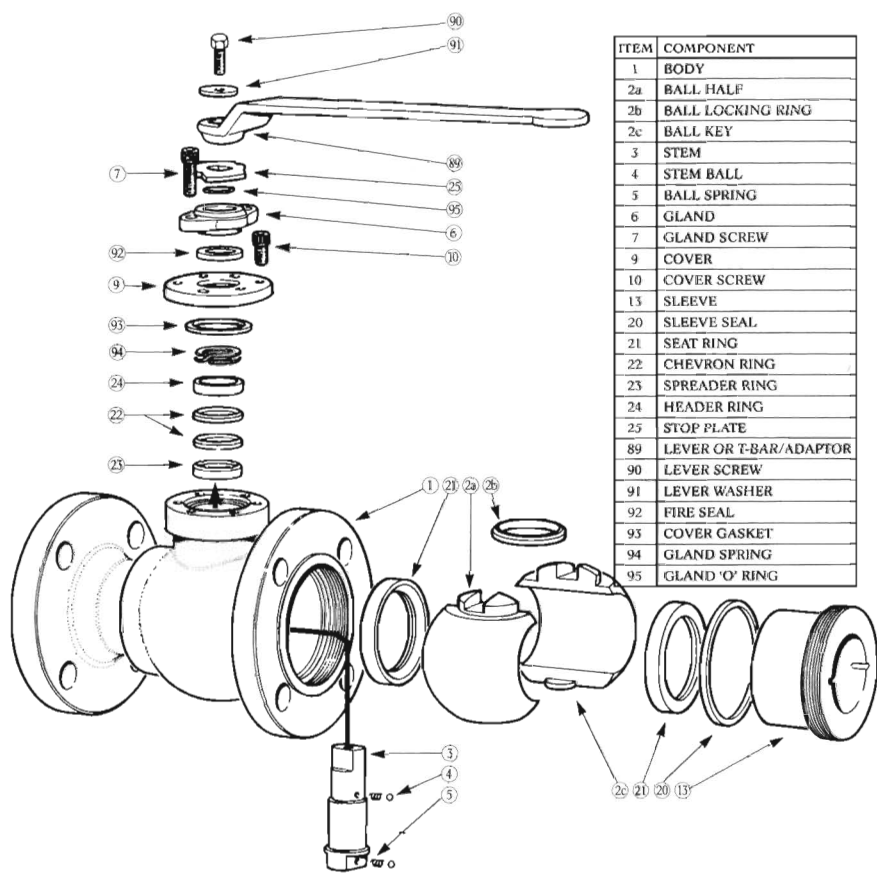


Figure 3. Fire test certified, floating full-bore ANSI class 150 and 300 ball valve.

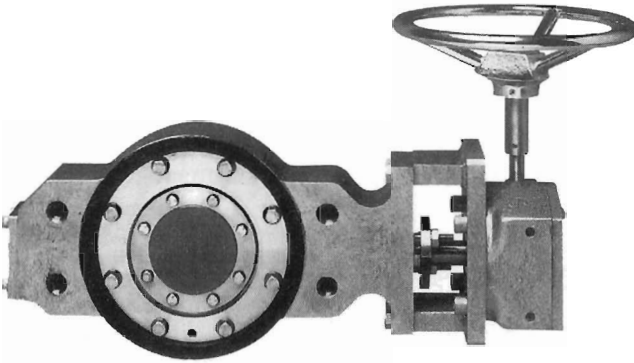


Figure 4. 150 mm wafer-type valve with stainless-steel laminated seal. Certified to BS 6755 Part 2.

 BUREAU VERITAS INDUSTRY DIVISION	APPROVAL CERTIFICATE		n° M7654/C	Page 1 / 1
	Ref. BV.ITA.: 40.96.7654			

FIRE SAFE TEST ACC. TO API 607/BS 6755	
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Client : OMB SPA	Inspection date: 04.11.96
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The undersigned ALFIO NICOLINI, Surveyor of BUREAU VERITAS, acting within the scope of the general conditions which regulate the interventions of our company, did attend at request of Messrs. OMB SPA CENATE SOTTO - BERGAMO to their works for the purpose of witnessing the Fire Safe Test according to API 607-1985 and BS 6755 PART 2-1987 on the following valve:

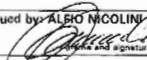
Valve Description

Ball Side Entry BSE Trunnion Valve 900 Lbs. Flanged Stock Finish
 size 8 x 4" Reduced Bore
 Material Body-Closure : ASTM A105N + ENP
 Material Seat : ASTM A182 F6a Class 2 + Nylon 12
 Material Stem : ASTM A106N + ENP
 Material Ball : ASTM A182 F6a Class 2

Manufactured and assembled in accordance with OMB drawing no. A900A1900

The valve was exposed to fire for 30 minutes, and during the test all parameters, as required by API 607 and BS6755 Part 2, were checked, recorded and found according to STD requirements. For further details of test arrangements and test results, see OMB Certificate no. 1569 herewith attached and duly endorsed.

ON THE BASIS OF THE RESULTS, THE ABOVE BALL VALVE SATISFACTORILY PASSED THE " FIRE SAFE TEST "

Date: 04.11.96	Issued by: ALFIO NICOLINI  Inspector Center: MILANO
	Diffusion: _____ KD Client: ☐ ☐

Typical fire-safe certification.

Table 3. Advantages of triple-offset segment valve over gate and plug valves

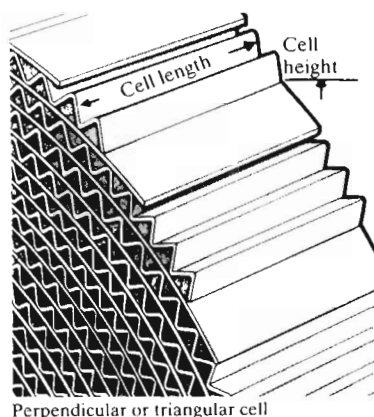
	Gate	Plug	Ball	Triple-offset segment
Sealing	Jam closures: leak becomes progressively worse with each cycle.		Pressure and/or mechanically assisted seating maintains seal.	Unique torque sealing.
Actuation	Manual: requires high torque to jam. Stem threads corrode and resist turning, often break in open or closed position.	High torque due to surface area, requires frequent takeup of sealing nut increasing torque. Impractical in gritty, caking, coking or gummy service.	Consistent torque. $\frac{1}{4}$ turn rotation. Stops must be set.	Low torque. $\frac{1}{4}$ turn rotation. No stops required. Seat provides mechanical stop.
Throttling	Not used	Impractical	Yes	Yes
Relative size	Very high, bulky and heavy. Automation very large.	Bulky. Automation very large.	Large. Automated package smaller than comparable gate or plug.	Light-weight. Most compact...
Maintenance costs	Very high, particularly in corrosive or erosive service due to stem freezing, wedge failure, and seat erosion. Requires frequent preventive maintenance (greasing, etc.).	High, particularly in corrosive or gritty service.	Medium.	Very low.
Labour intensity	HIGH Installation (heavy weight). Manual actuation (high torque). Preventive maintenance. Overall maintenance.	HIGH Installation (heavy weight). Manual actuation (high torque). Preventive maintenance.	MEDIUM Medium to heavyweight. Body needs to be split for maintenance.	VERY LOW Light-weight. Maintenance is simple.
Cost of ownership				
Purchase	High	High	Medium	Lowest
Installation	High	High	Lowest	Lowest
Operation	Highest	High	Low	Lowest
Maintenance	Highest	High	Low	Long
Life	Long	Long	Long	Long
Cost of actuation	Highest	High	Low	Lowest

Flame arresters prevent the propagation or spread of flames by absorbing and dissipating heat from the flames on one side of the arrester, so preventing the temperature of the gas or vapour on the opposite side rising to ignition point.

Ignited gas or vapour attempting to pass through a flame arrester is extinguished by the element assembly absorbing heat from the flame faster than it is developed.

Thus the temperature of the flame is lowered below its ignition point. To accomplish this, the area of metal surface of the arrester element must be sufficient to absorb the heat generated.

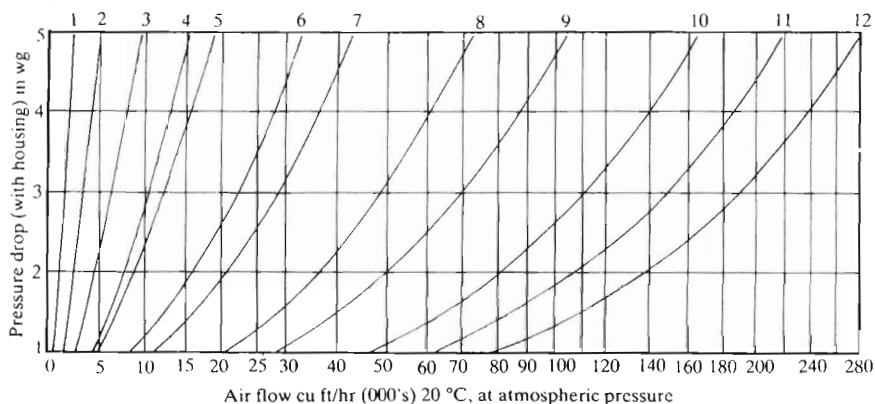
Flame arresters are used extensively in the chemical, petrochemical, petroleum, gas, marine, mining, aircraft and many other industries. They are



0.024 in cell height

Graph No.	Flame arrester size (nominal bore of pipe)	
	mm	in
1	19 to 25	¾ to 1
2	38	1½
3	50	2
4	63	2½
5	76	3
6	100	4
7	127	5
8	152	6
9	200	8
10	200	8
11	250	10
12	300 to 500	12 to 20

Graph No.



For other gases the equivalent air flow is given by multiplying the gas flow by the square root of its specific gravity.

Figure 5. Air flow through flame arrester elements in housings.

also used for purging gas mains, on the air intakes of internal combustion engines, and in many processes using solvents and gases.

Flame arresters are made in two basic forms: for fitting into pipelines, and onto the end of a pipe. The elements are made up of two strips of thin foil, one corrugated and the other plain. The strips are placed together and coiled into circular elements to give a spiral matrix of triangular cells and the whole assembly is fitted into an outer case.

Elements are made with various cell sizes and widths depending on the applications and the gas or vapour concerned. Elements are constructed either from cupro-nickel with a brass outer case, or stainless steel for both matrix and casing.

Flame arresters should be inspected frequently as part of routine plant maintenance and cleaned as necessary, and certainly if excessive pressure drop is experienced due to fouling of element cells. In the event of a flash-back,

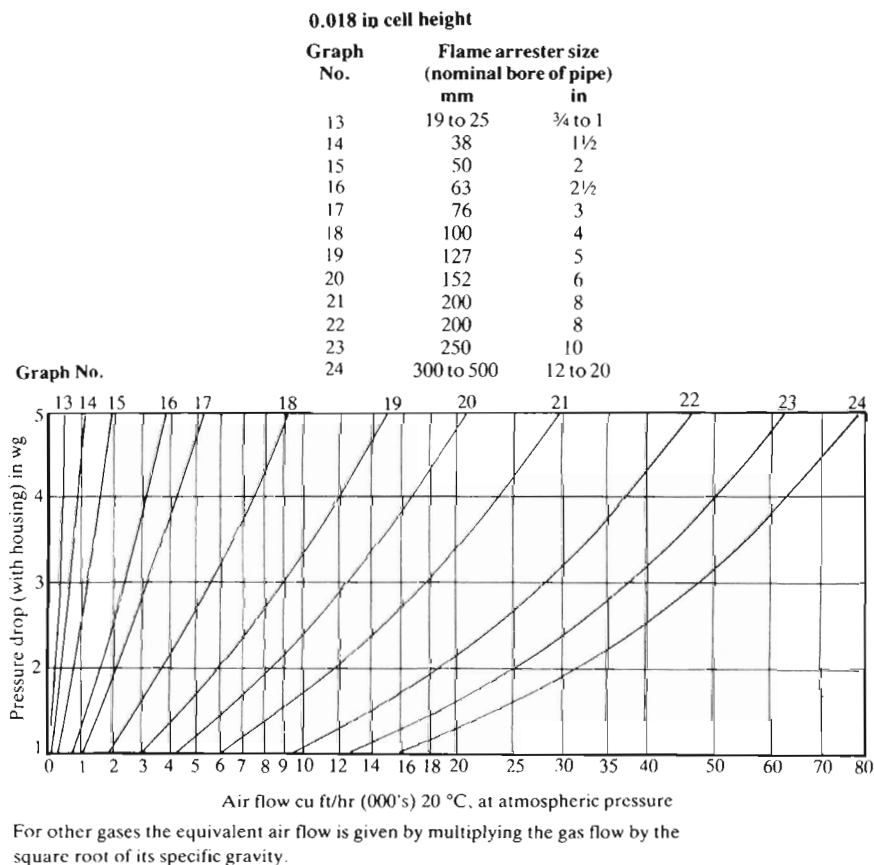
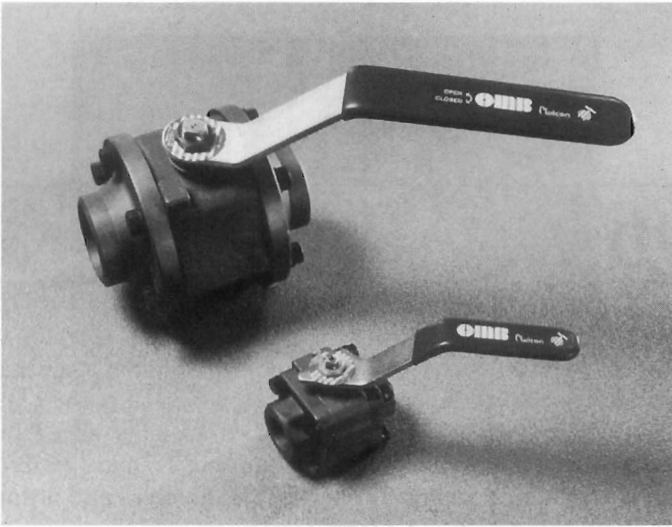


Figure 6. Air flow through flame arrester elements in housings.



Fire-safe ball valves to API 607 and BS 5146.

the flame arrester should be inspected immediately and if the element is damaged or distorted, a replacement should be fitted.

All flame arresters offer some resistance to flow defined as pressure drop. This value increases with the degree of hazard due to the longer cell sizes and smaller crimp required in manufacture of the arrester element to absorb heat or a specific application. To reduce the pressure drop to a minimum, elements have a straight-through cell construction and the element housing is increased to an area several times that of the nominal bore of the pipe.

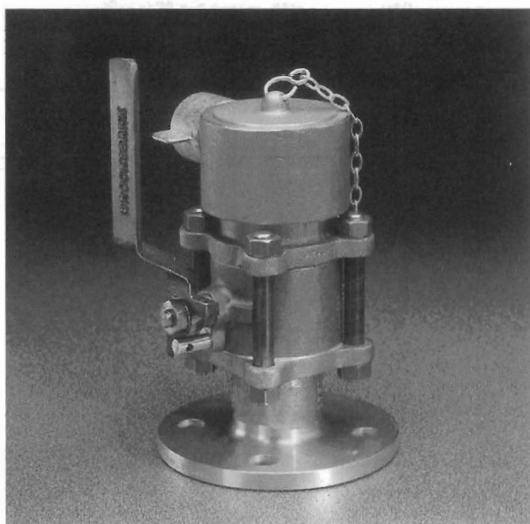
Graphs of pressure drop against airflow are shown for elements with a cell height of 0.61 mm (0.024 in) in Figure 5, and a cell height of 0.46 mm (0.018 in) in Figure 6.

Fire Hydrant Valves

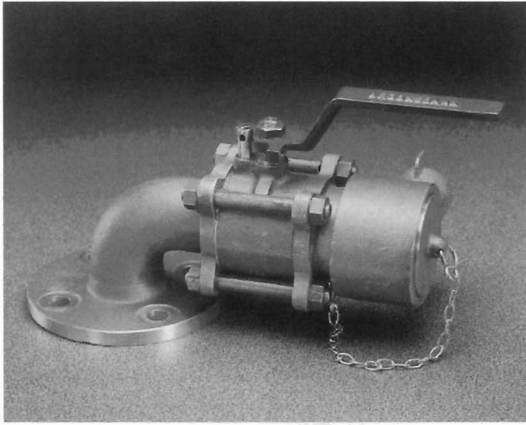
Fire-hydrant valves comprise straight-way and angle ball valves as well as screw-down gate or globe valves purpose-designed to meet the requirements of International Standards, e.g. BS 750:1984 which calls for a delivery of 34 l/s (450 gal/min) at a constant running pressure of 1.7 bar (25 lbf/in²). Ease of operation is also a most important feature of all valves and fittings used in fire-fighting.

The fixed-valve fire hydrant shown in Figure 1 conforms to BS750:1984 Type 2, screwed-down underground fire hydrant and exceeds the flow-delivery requirements of 1.7 bar.

The inlet mounting flange has been designed so that it is suitable for new and existing installations. The option of fixed or loose valve, either gunmetal or plastic outlet and the choice of packed gland or O-ring stem seal makes this type of fire-hydrant valve particularly versatile. A typical flow performance curve is shown in Table 1.



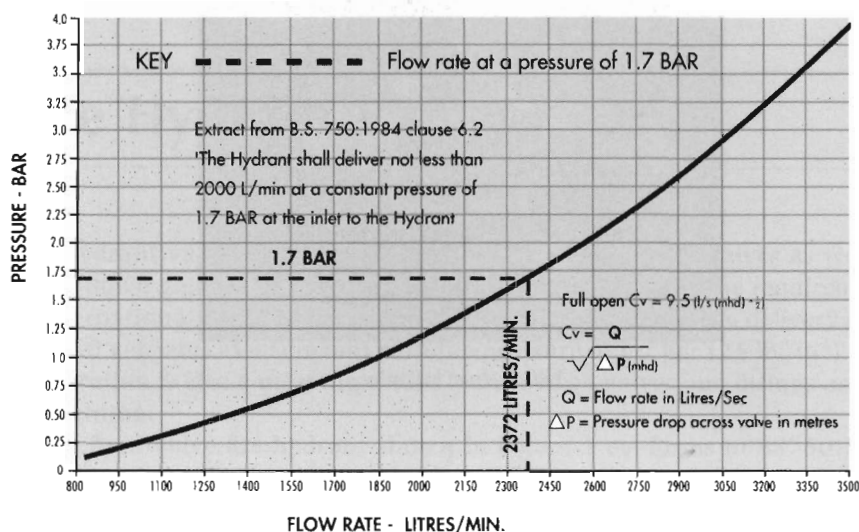
Straight-way hydrant ball valve.



Angle hydrant ball valve.



Figure 1. Multi-purpose fixed-valve fire hydrant.

Table 1. Typical performance curve**FLOW PERFORMANCE CURVE**

If the valve seat and O-seal to the body cover need to be replaced, it is important to ensure that the main has been depressurised before the valve is dismantled.

Depending on local conditions, fire-hydrant valves should be inspected at least every 3 months for:

1. physical damage to the surface box
2. rubble or silt in the chamber preventing access to the hydrant
3. loss of parts.

Every 12 months, the valve should be tested unless the visual inspection reveals any damage. The valve should be checked to see that it is free from leaks and operates correctly when fully open.

Fire landing valves

These valves are designed to meet the technical requirements of wet-rise systems in high-rise buildings or other structural installations where similar fire-protection systems are employed. They are typically manufactured with flanged inlet and instantaneous female outlet connections with metal-to-metal seating.



Figure 2. Standard Class F hose pressure regulator.

Pressure-regulating valves

Combined hydrant-stop and pressure-regulating valves are used for a wide range of fire-protection equipment suitable for refinery, chemical, petrochemical plants and offshore applications.

The Class 'F' valve shown in Figure 2 is a high-pressure regulator.

It is suitable for:

- fire mains systems in high-rise buildings
- high-pressure systems on oil-rig platforms and in oil refineries and chemical plants
- fixed monitors and hand lines where individual pressure requirements vary
- applications with high pressure drops caused by the length of water mains
- applications with low-pressure conditions produced by pump characteristics

This type of valve maintains a uniform fire-fighting pressure at every hydrant in a fire-protection system, irrespective of location.

The unit incorporates a spring-loaded 'balanced' pressure-reducing valve combined with a hydrant-stop valve. The stop-valve element is operated in exactly the same way as a conventional hydrant-stop valve.

The reducing-valve element is opened by the load applied to the pressure-adjusting spring and closed by the reduced pressure acting upon the underside of the low-pressure seal.

Under working conditions, the balance of these two forces determines the degree of valve opening required to maintain a steady outlet pressure.

Pressure control is achieved by a venturi section in the outlet flow area.

Valves of this type may also be fitted with a set-pressure override device which, when actuated, allows full opening of the valve without regulating the downstream pressure, thereby bringing it close to the available inlet pressure.

Marine Services

The main requirement in marine valves is full material compatibility with the fluid being handled, e.g. gunmetal or nickel–aluminium bronze being a normal choice for sea-water systems. Corrosion problems are often aggravated by the fact that many such valves, e.g. sea cocks, can remain open or closed for long periods. The type of valve used is largely immaterial provided it performs the required function, but ball valves are generally preferred. Table 1 lists some applications typical of naval vessels where the highest standards are normally specified.

Environmental pollution caused by a series of oil tanker accidents has resulted in a number of attempts to improve standards and legislation in the marine sector. Until legislation is formally established, there will be uncertainty about requirements for valves and users should always look to the highest standards for guidance.

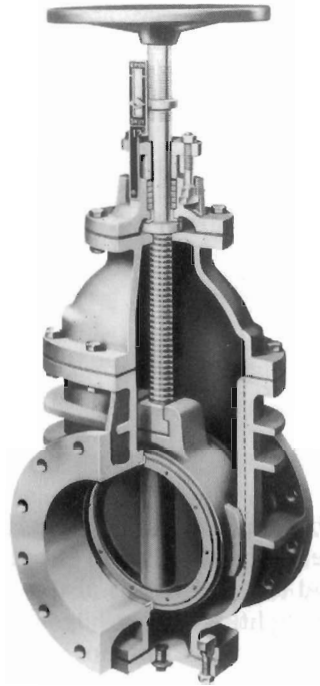


Lloyds and defence approved two-piece fire-safe ball valve.

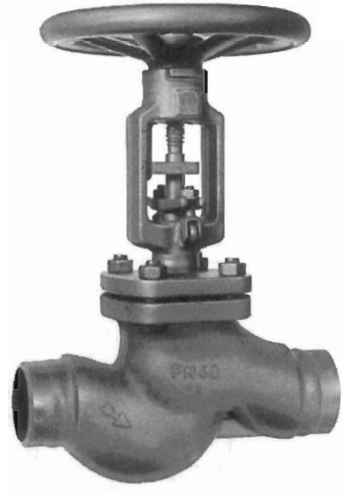
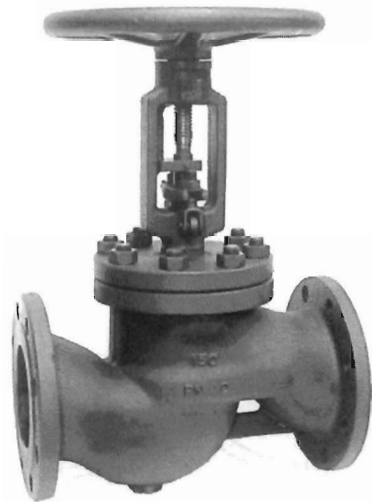
Table 1. Applications for naval vessels

Application and working pressure	Material	Valve type	Systems	Remarks
LP fluid services General 15.86 bar (230 lbf/in ²)	Gunmetal	Flanged	Sea-water Chilled water Fresh water Air Furnace fuel oil Diesel fuel oil Lubricating oil	May also be produced in non-magnetic materials.
LP fluid services General 15.86 bar (230 lbf/in ²) MP fluid services 34.47 bar (500 lbf/in ²)	Ni, Al, bronze	Screwed Female BSP	Sea-water Chilled water Air Furnace fuel oil Diesel fuel oil Lubricating oil	
LP fluid services General 15.86 bar (230 lbf/in ²)	Aluminium	Screwed Female BSP	Some cooling systems Air systems Fuel systems Lubricating oil Where weight is at a premium	These valves are non-magnetic.
LP fuel systems 15.86 bar (230 lbf/in ²) Fire-safe	Carbon steel	Flanged	Furnace fuel oil Diesel fuel oil Lubricating oil	
LP fuel systems 15.86 bar (230 lbf/in ²) Fire-safe	Stainless steel	Flanged	Helicopter fuelling systems and some gas turbine fuel systems where scrupulous cleanliness is required. Sea-water displaced fuel systems where any possible corrosion risk is secondary to a fire-safe requirement	May also be produced in non-magnetic materials.
HP fuel systems 62.05 bar (900 lbf/in ²) Fire-safe	Carbon steel	Flanged	Furnace fuel oil or diesel fuel oil supply to certain types of steam boilers	Fire-safe valves.
MP fuel systems 27.58 bar (400 lbf/in ²) Fire-safe	Carbon steel	Flanged ANSI 300	Fuel supply for certain steam atomisation type of boilers. Refrigeration gas for refrigerant systems	
HP hydraulic fluids 62.05 bar (900 lbf/in ²)	Carbon steel	Screwed Female BSP or unified	Missile handling hydraulics HP air Gun turret, general service hydraulics Propellor pitch control	Fire-safe valves.
Actuators 5.25 bar (80 lbf/in ²) 8.27 bar (120 lbf/in ²)	Steel and aluminium			

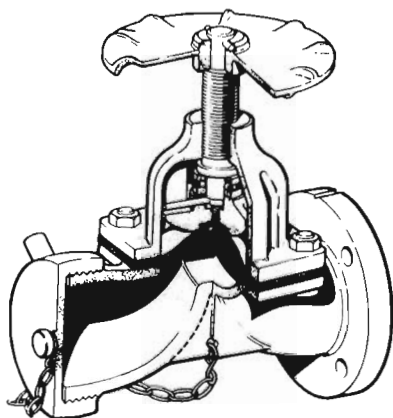
LP, low pressure; MP, medium pressure; HP, high pressure.



Marine tanker wedge-gate valve.



Globe valves with non-turning stems for certain marine service applications.



Marine tank cleaning valve.

When loading and discharging liquids and gases through port installations, environmental requirements insist that valves must be leak-tight to avoid leakage into harbours and docks. There is also a requirement for ancillary services such as water, fire-fighting, boiler, bilge and tank cleaning, with the latter having leak-tight requirements. Gas movements in offshore areas are by pipeline and tanker and both safety and environmental requirements call for leak-tight operations.

Vacuum Services

The main types of valves used for vacuum services are butterfly, diaphragm, globe, gate and ball valves. Typical forms of diaphragm vacuum valves are shown in Figure 1.

The factor which limits the pressure at which a diaphragm valve can be used is that a large area of elastomer (usually nitrile rubber) is exposed to the process. At low pressures, the molecules trapped on the surface of the elastomer are given off, limiting the ultimate pressure which can be obtained. Although materials such as Viton and PTFE, which have lower outgassing rates, can be used, it is more usual to employ different types of valve for low-pressure applications.

In medium to high applications where automatic or remote control is required, magnetically-operated valves can be used. These valves have nitrile rubber washer seals which are kept open under spring load. Air admittance valves, which automatically open when switched off, are also available. As a general rule, reinforced diaphragms are used in valves for vacuum services.

The ball valve shown in Figure 2 is a soft-seated bi-directional sealing valve suitable for vacuum down to 2×10^{-2} torr. The standard seat-ring material is virgin PTFE or UHMW Polyethylene.

High-performance butterfly valves using the wafer-type sealing principle are capable of vacuum tight sealing up to 2.264×10^{-5} bar (2×10^{-2} torr) (Figure 3).

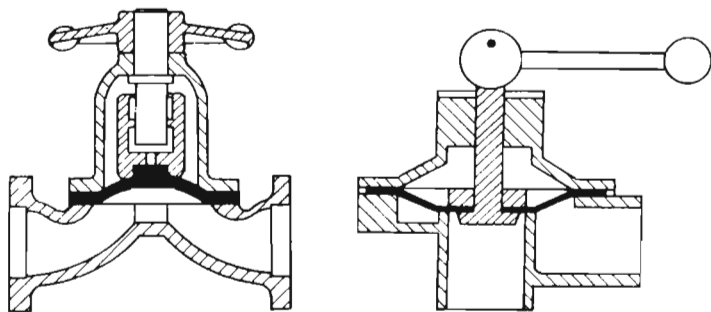
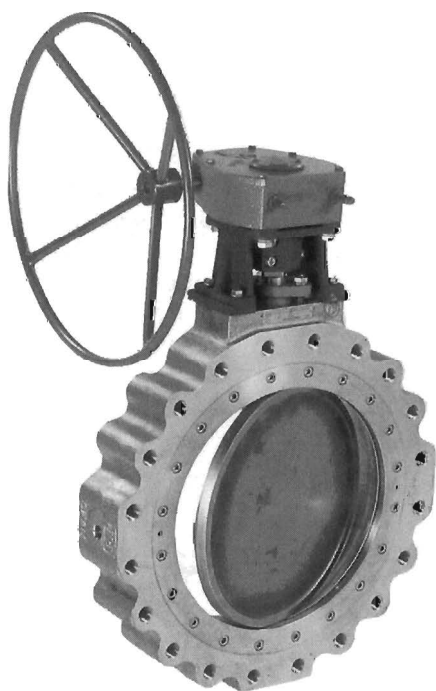


Figure 1.



Figure 2. Reduced-bore ball valve for vacuum down to 2×10^{-2} torr.



High-performance butterfly valve.

A simple single-piece, flexible polymeric seat that is pressure-energised provides positive shut-off. The seat is designed to seal effectively regardless of direction of flow. Butterfly valves are a useful type for vacuum services.

Vacuum seal-off valves

The vacuum seal-off valve generally offers a highly reliable means of evacuating a vacuum space and a high degree of resistance to accidental opening or tampering.

These valves are typically used in conjunction with a 'valve operator' which opens and closes the seal-off valve (Figure 4).

The valve operator is installed onto the seal-off valve and vacuum applied to the side port. The valve-operator stem is inserted, and the inner valve unthreaded and withdrawn into the valve operator. This provides maximum conductance for evacuation.

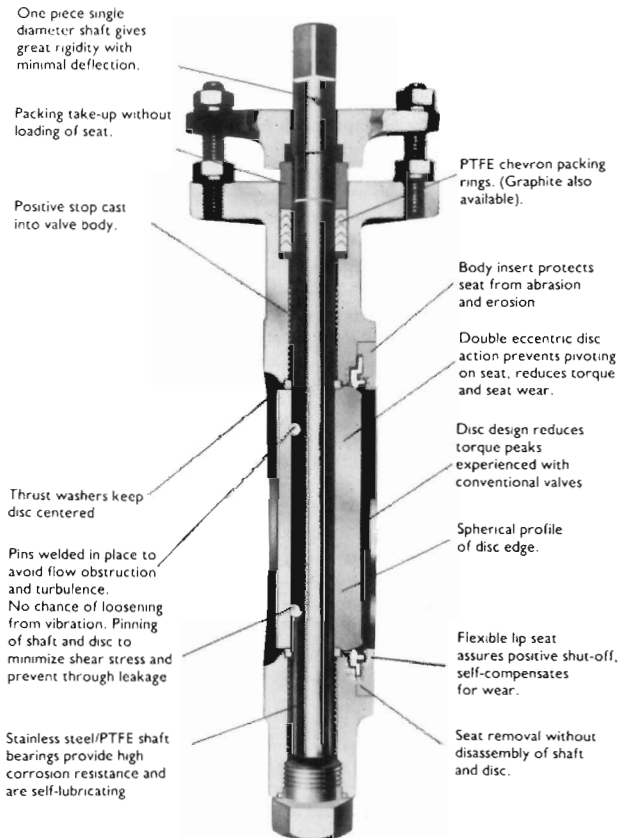


Figure 3. Butterfly valve of 'wafer-sphere' design for vacuum service.



Figure 4. Vacuum seal-off valve (left) and valve operator (right).

Table 1. Valves for vacuum services

Vacuum	Valve type	Remarks
Rough to medium	Diaphragm	Outgassing from elastomeric diaphragm limits the ultimate pressure which can be obtained in the system.
	Globe	With metallic bellows bonnet seal.
	Ball	Generally more suitable than other types.
Medium to high	Ball	More precisely machined than for medium to rough service.
	Plate	May be preferred to ball or diaphragm valve for services down to 10^{-7} torr.
	Quarter-turn swing valve	As pipeline valve, or incorporated in a pumping stack.
	Gate	Higher conductivity than quarter-turn swing valve.
	Baffle valves	Baffles associated with an isolating valve for isolating a working vapour diffusion pump when the system is let up to atmospheric pressure.
Very high	Right-angle plate valve	With non-elastomeric seals.
	Quarter-swing valve	
	Ball valve	
	Gate valve	
	Baffle valve	
Ultra-high	High conductivity type	Special designs and constructions.

Following the evacuation process, the handle is re-inserted, the inner seal tightened into place to establish the seat seal and the valve operator removed.

The valves described in Table 1 are just a typical basic selection for use in vacuum conditions.

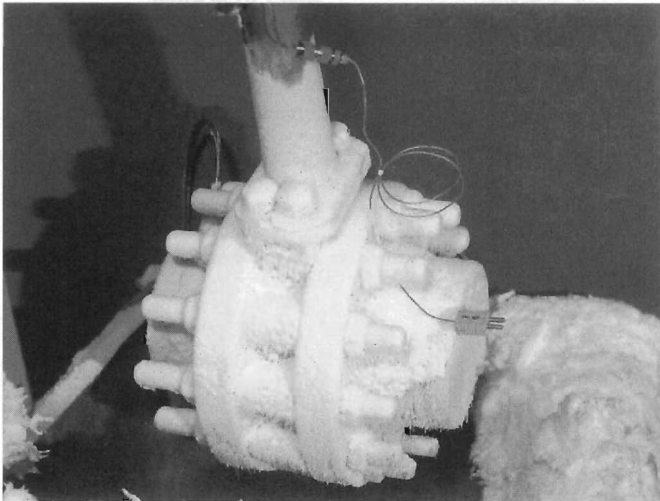
Cryogenic Valves

Cryogenics is the branch of physics dealing with processes and materials at very low temperatures, normally below -101°C (-239°F). The need to handle very low-temperature liquids and gases is a commonplace requirement with fluids such as oxygen, liquid hydrogen and nitrogen.

Valves that can provide tight shut-off for isolation or modulation for control requirements at temperatures down to -196°C (-410°F) are an essential part of process plant.

Valves that are lightweight in construction are better suited to cryogenic use since the valve mass which must be cooled down from ambient to cryogenic temperatures on start-up is much reduced. Also, the lower conductivity of lighter weight valves can assist in reducing heat influx which can occur in heavier design styles of valve.

The use of valves in low-temperature and cryogenic conditions presents an array of problems.



Cryogenic triple-offset butterfly valve.

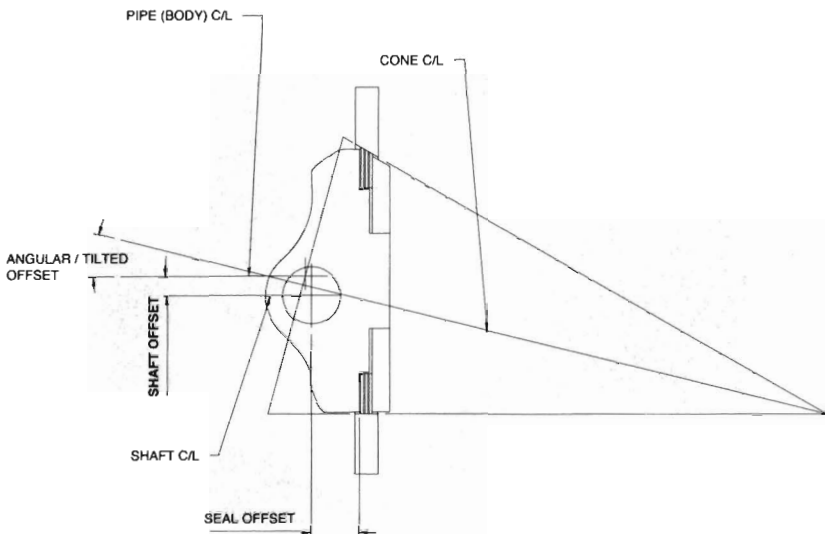
The definition as to what temperature is to be deemed cryogenic as opposed to low temperature is to some degree arbitrary, but it is generally accepted that low temperature refers to temperatures below minimum ambient (about -30°C) down to -100°C . In considering the types of valves to be used in such services, it should be borne in mind that leakage rates which may be acceptable in conventional applications may not be acceptable at cryogenic temperatures.

Any leakage of fluids, whether to the downstream side of a valve or to the atmosphere through gland seals or gaskets, will result in instant freezing with probable adverse effects on the performance of valves and associated equipment.

Such leakage at the valve stem would lead to increased localised icing, which could culminate in seizure of the valve-operating mechanism.

Valve types used for handling liquid oxygen, nitrogen, methane, natural gas, ammonia, etc., at cryogenic temperatures may be of ball, gate, globe, butterfly or check type, with an increasing preference for high performance and true-zero leakage as defined by API598. All are special designs, normally identified by having extended bonnets to position the valve-stem seals away from the cold source. This extension can also serve the purpose of providing an insulation space between the pipeline and the lever or handwheel operating the valve.

Valves for cryogenic services may range in size from 3 to 2240 mm ($1/8$ to 88 in), with pressure ratings from ultra-high vacuum up to 700 bar (10,000 lbf/in²), capable of working down to -254°C (-425°F).



Triple-offset geometry.

Construction may be in aluminium, brass, bronze, monel, incoloy, stainless steel and zirconium, rendered antistatic either by a graphite gland assembly or external bonding. On ball valves, specially designed seats are required to minimise the effect of differential expansion by reducing seal volume to a minimum. Seals are normally made in PTFE.

Valve-stem seals must be of a type not requiring lubrication (lubricants would freeze; also they could not be tolerated in oxygen systems). Seals used are normally graphite, TFE or PTFE. Pressure is normally equalised between the valve body and cavity within the extended bonnet. All valves for cryogenic services need to be cleaned, degreased and finally assembled in a clean room.

Low-temperature valves

Valves designed for low-temperature but not cryogenic services may follow a similar form, but not necessarily, the same extended bonnet. Valve bodies may also be in carbon steel instead of stainless steel. Suitable carbon steels are available for services down to -73°C (-100°F).

Triple offset

A typical high-pressure performance valve for cryogenic applications is the triple-offset butterfly valve. This type of valve has become increasingly popular over the single- and double-offset design (Figure 1).

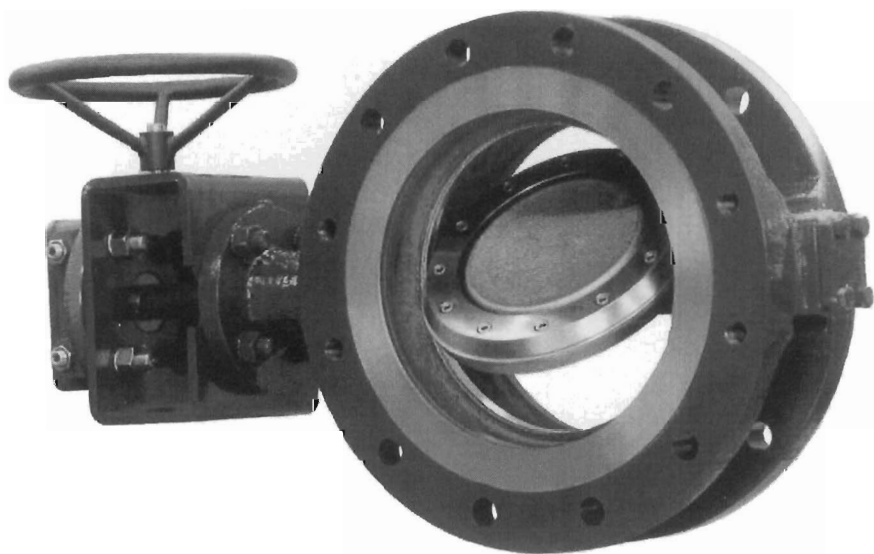


Figure 1. Triple-offset cryogenic butterfly valve.

The shaft offsets are created by designing the valve with the shaft located behind the centre-line of the sealing surface and slightly to one side of the pipe centre-line. The function of these offsets is to reduce the rubbing and thus the wear between the seat and seal to approximately 20° of travel as well as to eliminate all seat to seal rubbing throughout the valve's entire 90° of rotation (Figure 2).

Typically, triple-offset valves (TOSV) of this type used for cryogenic duties have a resilient stainless-steel ring installed in the disk assembly and it is this that provides a true 'zero-leakage' seal.

The seal and seat-contact surface is 'cone-in-cone' where both cones are inclined and the angle of contact between the seal and seat generates a slight wedging effect that flexes and radially compresses the disc-seal ring. The valve is able to shut-off completely, regardless of the direction of flow or line pressure.

Ball valves

Double-seal, reduced- and full-bore ball valves are used for LPG, LNG, thermal fluids and other cold applications including oxygen and nitrogen (Figure 3).

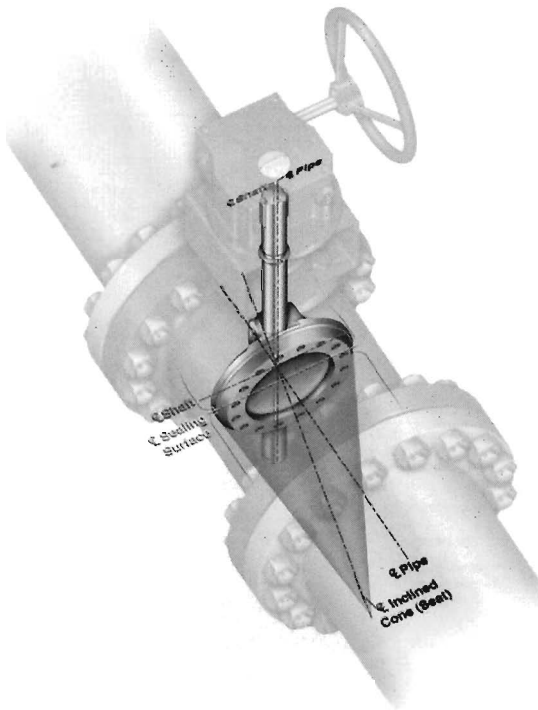


Figure 2. Triple-offset inclined-cone configuration.

Valves of this type incorporate a vapour space of sufficient height to allow gasification in the area below the gland. This ensures that the gland packing remains at near ambient temperature.

When standard valves are fitted with actuators, they should be installed with extensions in the vertical position.

The LNG market has imposed its own demands on valves as the profile of valve usage changes. Modern LPG operations have meant that pipe and valve sizes have increased dramatically, with a much larger percentage being fitted with actuators for remote operation. These factors have led to increasing demand for the quarter-turn ball and soft-seated double-eccentric butterfly valve types which offer easier actuation and lower size/weight ratio combined with high performance in terms of shut-off capability and lifetime durability.

Gate and globe valves are more difficult and costly to actuate and become less economical as pipeline size increases. Such valve types tend to be restricted to use on smaller pipe sizes and applications where remote operation is not required.

High purity

Ultra-clean process technologies require preservation of product cleanliness throughout the gas-distribution system from cryogen storage to point of use delivery.

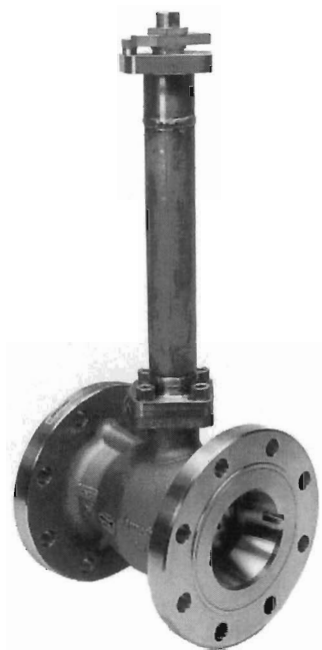
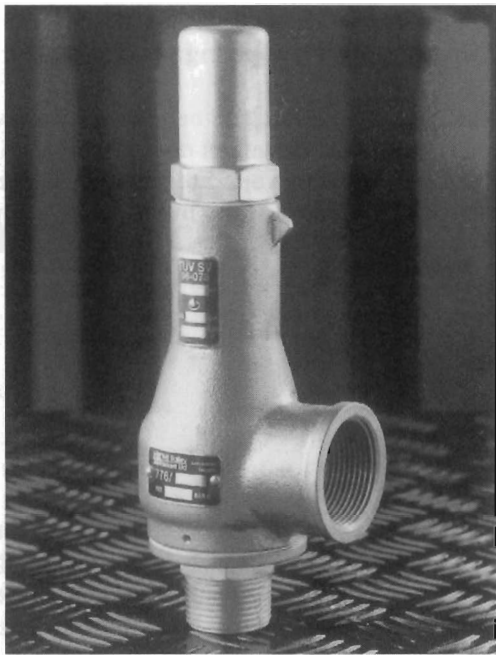


Figure 3. Double-sealed reduced- and full-bore ball valve for cold applications.



Safety valve for cryogenic duty.

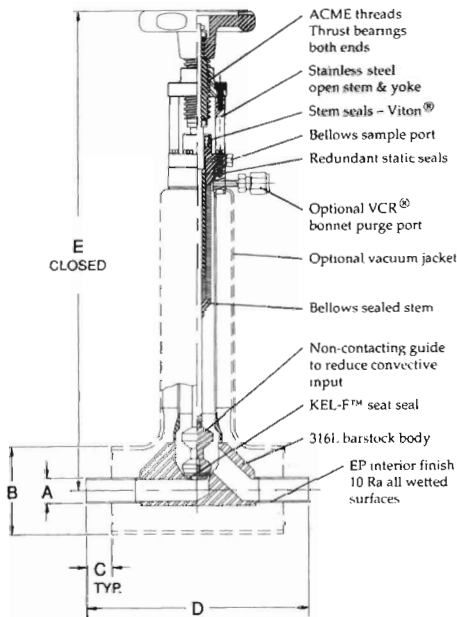


Figure 4. Ultra-high purity cryogenic clean valve.

Ultra-high purity gas valves (Figure 4) are used in a variety of applications including dewars, vacuum-insulated piping systems, trailers, tanks and cold boxes.

Typically, the interior surfaces and the bellows which seal the valve stem are electropolished. This type of valve has no contacting metal surfaces and may be manually or pneumatically operated with a safety return actuator which returns the valve to fail-safe position in the event of supply failure.

Nuclear Services

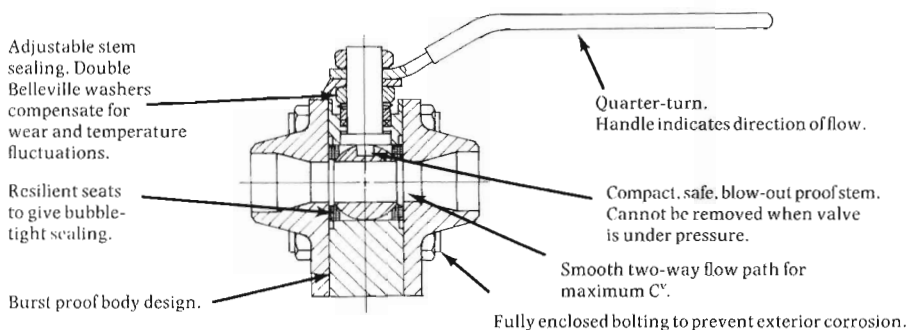
Nuclear valve glands

Most types of valves can be found in a variety of applications in nuclear services. These include: nuclear power plant, primary, secondary and auxiliary systems, fuel cycle dangerous fluid (radioactivity), fundamental research, experimental reactors, test loops, steam-supply systems for the propulsion of nuclear submarines and aircraft carriers, and a variety of other applications from saturated or superheated steam to dangerous or corrosive fluids via special liquids or gasses.

Valve types include: wedge-gate valves, gland or bellows globe valves, plug and ball valves, check valves and instrumentation valves. Many of these products can be fitted with actuators. Other valves for special service include high-performance butterfly valves with PTFE, metallic or elastomer seats and pressure-relief valves for both primary and secondary systems.

Pressure-balanced safety valves are used extensively for pressurised water reactor primary service and dual-function valves for automatic or emergency override operation for boiling water reactors.

Pressure-relief valves for secondary steam systems include power-operated pressure-relief valves for two-phase flow, large-orifice, pilot-operated and



Typical ball-valve design for nuclear service.

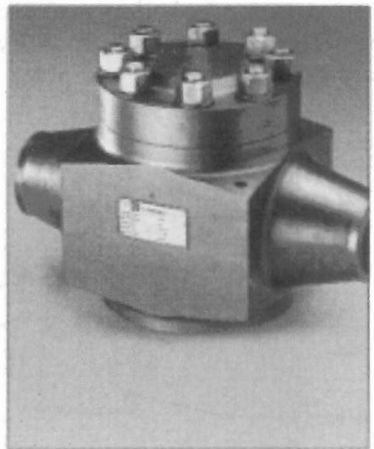
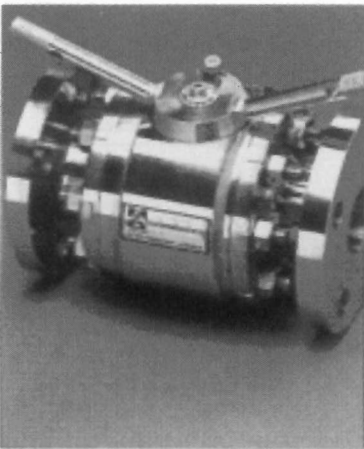
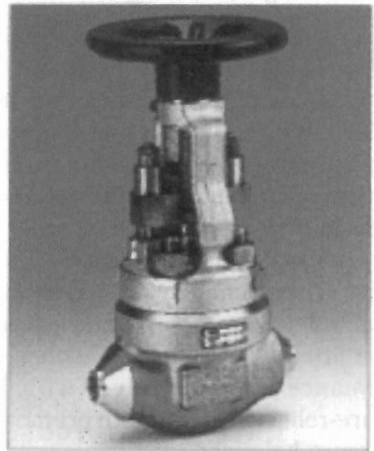
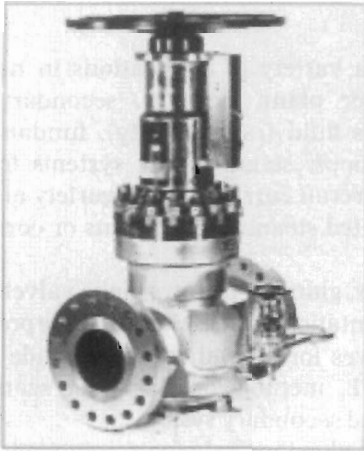
spring-loaded pressure-relief valves for protection of moisture-separator reheaters and pressure-relief valves for auxiliary fluid service.

Valve position indicator systems provide direct, continuous, remote indication of valve-stem position and permits positive monitoring of pressure-relief valves in nuclear environments.

Typical safety and pressure-relief valves are shown in Figure 1.

Electrically powered, rotary-actuated clamp valves are used successfully in nuclear power plants to provide full-flow on-off or throttling service (Figure 2).

The system includes an air motor that automatically operates the valve to the fully closed (or open) position on loss of electric power.



Isolating and regulating nuclear valves.

The very highest integrity is an obvious requirement of valve glands in nuclear plant, demanding satisfactory performance in the following areas:

- (i) mechanical: deformation, resistance and recovery
- (ii) dynamic: behaviour under spindle movement, under pressure and at elevated temperatures
- (iii) corrosion: elimination of valve steam corrosion.

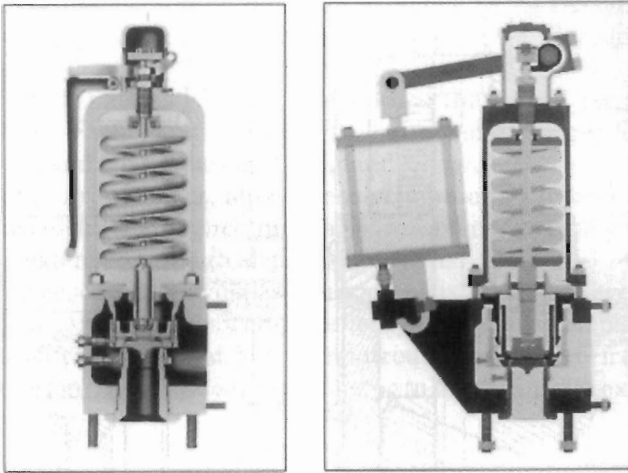


Figure 1. Pressure-relief and safety valves for nuclear service.



Figure 2. Electric-powered rotary-actuated clamp valve.

The main fluids to be handled are demineralised water, saturated steam (BWR plants) and borated water (PWR plants); but with possibly 1500 pieces of equipment in each division of a nuclear power plant, the multiplicity of problems involved can be immense. Desirable features of a nuclear valve gland are shown in Figure 3.

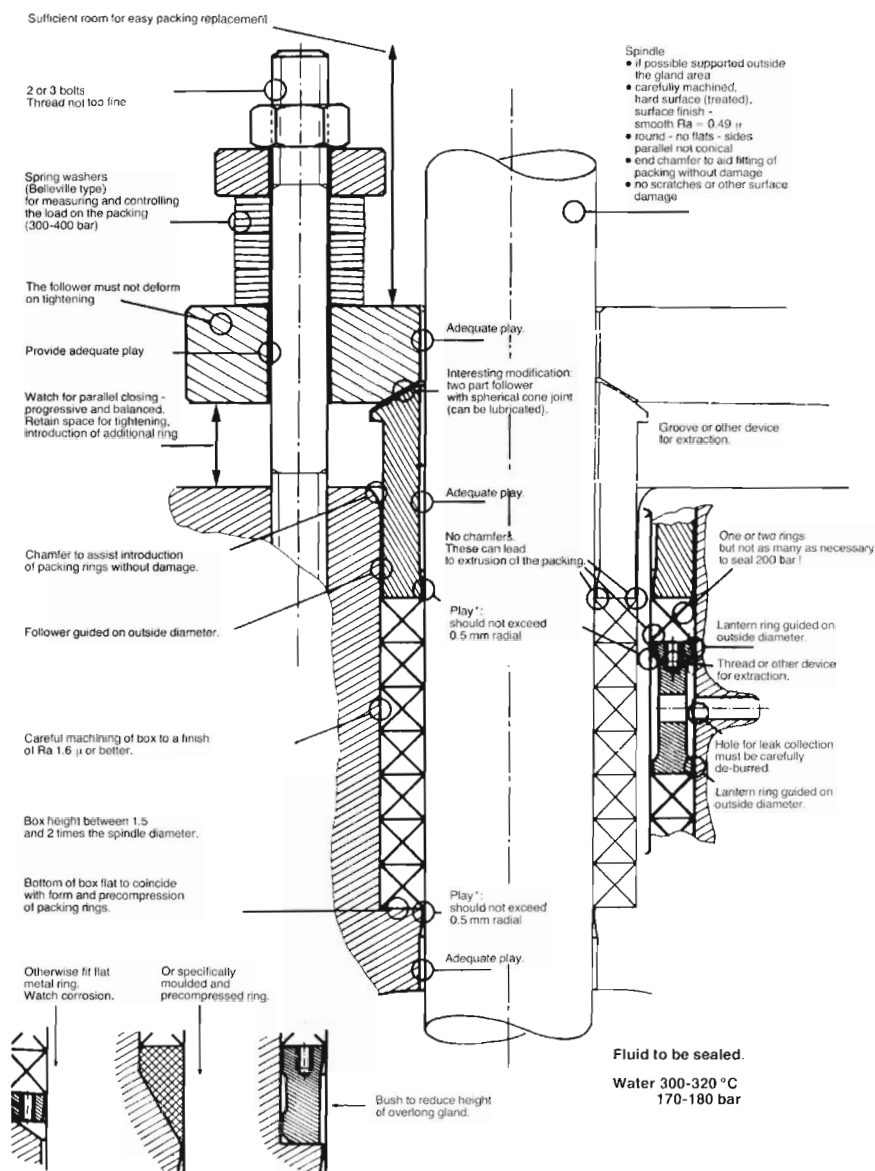


Figure 3. Desirable features of a nuclear valve gland.

The gland material now generally considered the most suitable in application is nuclear-purity expanded graphite (plus corrosion inhibition, if specifically called for). This is readily formed in mechanically sound rings from tape to provide the necessary resilience and deformability to behave as an efficient seal. To accommodate extrusion of the packing, expanded-graphite rings are normally combined with plaited rings at the top and bottom of the gland—a logical choice here for high-temperature working being pure graphite/asbestos braided packing which can also incorporate a corrosion inhibitor and a suitable proportion of anodes material to act as a sacrificial anode. An alternating arrangement of graphite fibre and expanded-graphite rings does not produce as satisfactory a seal.

The braided packing rings serve to eliminate the risk of extrusion of the expanded graphite where radial play at the bottom of the box exceeds 0.5 and 0.33 mm (0.02 and 0.01 in) around the gland follower.

This packing arrangement, after extensive laboratory testing by EDF, has been used successfully in French nuclear power plants for many years and is rapidly being extended through all primary circuits.

Should it be necessary to avoid asbestos products, rendering even wet-spun, dust-free product unacceptable, a graphite fibre packing can be employed for anti-extrusion purposes. This would, however, introduce problems of fragility and of potential corrosion risks (the stem alloys having to be chosen with extreme care).

Gland dimensions

The dimensional relationship between stem, box and packing rings is of prime importance. Interference between ring and stem with play between ring and box is to be avoided as leading to high stem torque and poor sealing. It is preferable to begin with a tight fit to the box and a small clearance to the stem, 0.1 mm (0.0004 in).

Surface finish is important, particularly on the valve stem, to realise minimum packing wear and low operating torque. Recommended values are $R_a = 0.4 \text{ m}$ for the stem and $R_a = 1.6 \text{ m}$ for the gland surface.

Spring discs

The maintenance of a leak-free seal is directly dependent on the maintenance of an adequate loading on the gland packing. Spring discs (Belleville-type washers) can compensate for loss of loading due to:

- relaxation of the packing, very slight for expanded graphite, of the order of 4% at 350 bar (5000 lbf/in²)
- wear
- differential expansions
- temperature variations

The introduction of spring discs also assist the precision with which gland loadings can be determined (i.e. by height reduction of disc). The cost of suitable spring discs is small in relation to the advantages they bring.

Gland geometry

The depth of a valve gland should be 1.5 to 2 times the stem diameter. A greater depth serves no purpose. Many glands are too deep. Should the number of packing rings exceed six to seven, transmission of gland loading becomes very uneven and stem torque increases disproportionately. Tests made by the EDF showed that an increased number of rings/depth of gland could result in increased leakage.

Recommendations for stem diameter/ring sizes are given in Table 1.

Causes of leakage

The initial cause of leakage developing is not always apparent, particularly as one fault can lead to another. Experience has indicated that, in order of seriousness, likely causes are:

- (i) The use of braided packings that lose volume too readily and harden in use.
- (ii) Damage to packing rings during fitting.
- (iii) Bad meeting of ring ends where cut rings are used.
- (iv) Incorrect disposition of ring joints, forming leak path.
- (v) Insufficient gland loading.
- (vi) Poor support to stem.
- (vii) Poor ability of packing to withstand thermal shock.
- (viii) Reduction in gland loading owing to packing relaxation, packing wear/volume loss.
- (ix) Incorrect dimensional tolerances between stem/packing box.
- (x) Gland too deep.
- (xi) Stem-surface finish of low order.

Table 1. Recommendations for stem diameter/ring size

Stem diameter (mm)	Ring square section (mm)
10	4
20	6
30	8
40	8–10
50	10
60	12

- (xii) Play at bottom and top of box too great.
- (xiii) Corrosion of stem and abrasion of packing during stem operation.
- (xiv) Too many rings above lantern.

Corrective actions

The following are points to observe, not necessarily in order of significance. Each one is important in achieving satisfactory gland performance.

1. Correct gland design.
2. Entry to facilitate fitting of packing rings.
3. Optimum gland depth.
4. Correct surface finishes.
5. Adequate capacity for gland loading/adjustment.
6. Spring discs to compensate for wear.
7. Good stem support.
8. Use of expanded-graphite sealing rings particularly recommended (expanded graphite is permanent, resilient, has better relaxation, maintains its volume and withstands thermal shock).
9. Correct dimensional tolerances between packings and gland.
10. Correct fitting and loading of gland before service operation.

Nuclear power piping

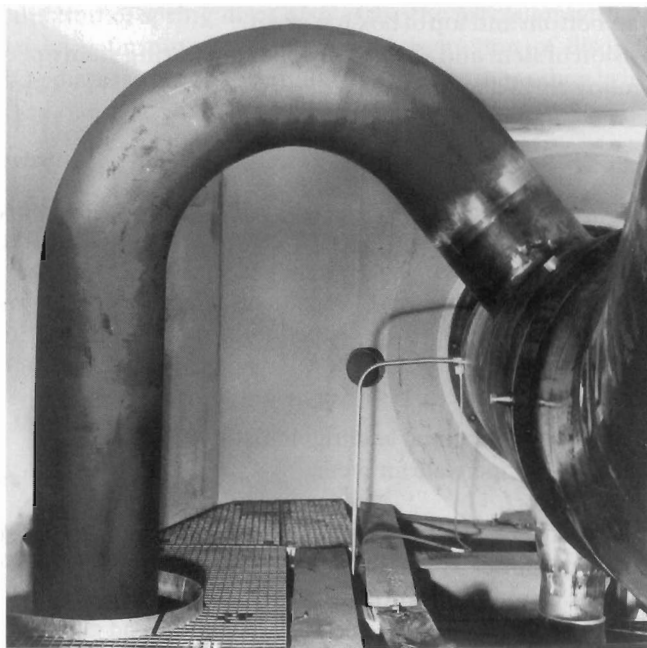
Special safety standards are required for the design of piping in nuclear plants. For a proposed British PWR plant, the design is based on the American Standardised Unit Nuclear Power Plant System (SNUPPS).

The safety-related piping are those pipelines associated with the safe operation and shut-down of the reactor. These can vary from the main coolant loop pipelines to the coolant high-pressure injection systems, and the coolant pipelines necessary for heat removal from the reactor.

In a typical SNUPPS design, the length of safety-related piping can be approximately 17,100 m (56,090 ft) out of a total length of piping of about 90,000 m (295,200 ft). The design code used is the ASME Boiler and Pressure Vessel Code Section III and the safety-related piping are class 1, 2 and 3 pipelines. The ANSI B31.1 Power Piping Code is also used for other piping within the safety-related buildings, which consist of the reactor, auxiliary, control, fuel and diesel buildings.

Piping systems are designed to withstand the dead-weight of the pipe and contents and, where applicable, to withstand temperature changes and pressures up to 197 bar (2856 lbf/in²). Design for earthquake and pipe-break conditions has also to be taken into consideration.

A PWR plant is to be designed to withstand an earthquake at a level of 0.25 *g* free field. The Safe Shutdown Earthquake (SSE) is the level of earthquake at



Seamless pipe bend of the material $\times 10$ CrNiNb 18 9 (1.4550) within a piping system in a nuclear power station. Pipe dimensions: 348 mm inside diameter, 40 mm wall thickness; bending radius $R=1.5 \times OD$.

which the station is designed to be shut down safely, with continued capacity for heat removal from the reactor core, but not necessarily to have the ability to be started up again. The Operational Shutdown Earthquake (OSE) is a level set which, if exceeded, will result in the initiation of a controlled shut-down of the station. This level of earthquake has been set as one fifth of the SSE. As at the set level of OSE, the piping design is still covered by the analysis carried out under SSE loading, the only additional piping analysis required is in the detailed fatigue analysis required for class 1 piping. Sample analyses on selected class 1 pipes will be carried out to determine the effect of the additional cycles of OSE events on fatigue usage factor.

Pipe-break conditions are considered in high-energy pipelines containing fluid at a temperature above 95°C (203°F), and/or at a pressure exceeding 19 bar (275 lbf/in^2). Although piping is designed not to break, in order to ensure the safety of a PWR station under all foreseen circumstances, there is a requirement that breaks in high-energy pipelines arising from unanticipated events are considered. Piping which falls into this category includes the main reactor coolant piping as well as subsidiary piping systems.

Three main effects should be considered when designing for pipe-break conditions:

- (i) Main-loop piping break: pipes connected to the main-loop piping are required to continue to function following a loop break. These have to be designed to withstand movements imparted to them and any pressure transients involved caused by the sudden efflux of fluid from the broken loop.
- (ii) Pipe whip: caused by the sudden release of fluid when high-energy pipes break. It is a safety requirement that the whipping pipe does not cause the failure of other safety-related equipment in the vicinity. This is carried out either by segregating the system from other safety-related systems by walls or distance, or by enclosing the pipes in energy-absorbing devices to catch the pipe and absorb the energy in a controlled manner.
- (iii) Jet impingement: after a high-energy pipe break the high-velocity jet efflux released can cause damage to surrounding structures. Adjacent pipelines will therefore be analysed for the effect of jet impingement.

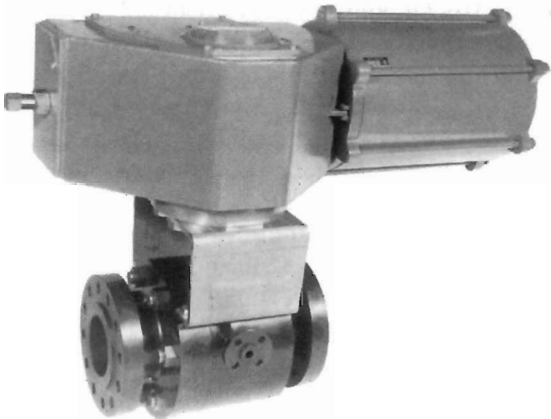
Other inputs will include water or steam hammer where these are likely to occur, detailed system transients for class 1 analysis, and general vibration caused by pumps where this can be identified as a likely occurrence.

High Pressure Services

High pressure can be so classified if it is in excess of 140 bar (2000 lb/in²g). Some typical high-pressure applications are shown in Table 1.

The units, symbols and conversion methods used in pressure measurement are as follows:

Unit/Symbol	Unit
lbf/in ²	pound force per square inch
N/in ²	newton per square metre
Pa	pascal
kgf/cm ²	kilogram force per square centimetre (technical atmosphere)
bar	bar = 10 ⁵ N/m ³
Atm.G.	standard atmosphere
1 kgf/cm ² = 14.223 lbf/in ² (AT)	1 technical atmosphere
1 N/in ² = 1.45038 × 10 ⁻⁴ lbf/in ²	1 pascal



High-pressure metal-seated ball valve for severe service.

In common with most other pipe systems, high-pressure systems require valves to isolate and control flow as well as to vent, drain, check and relieve the pressured medium. In all cases, the basic function of the valve is the same as any other standard pattern valve, the main differences being:

- (a) wall thickness
- (b) selection of materials
- (c) greater care in heat treatment and condition of materials
- (d) an extremely high standard of non-destructive testing and quality control
- (e) finer limits and fits, and surface finishes
- (f) closing members are more precisely formed
- (g) high manufacturing precision
- (h) each valve should have stress checks against application.

High-pressure valves

High-pressure valves fall into specific categories. All valves used on hydraulic circuits, for example, are high-pressure types, typically designed and constructed to accommodate working pressures of 140 bar (2000 lbf/in²), or higher in other specialised systems (e.g. aircraft hydraulics).

Table 1. Typical high-pressure applications

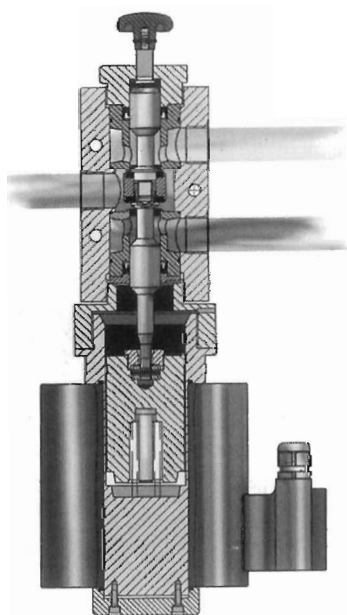
Pressure	Medium	Application
140 bar (2000 lb/in ² g)	Oil	Hydraulic systems
	Hydrogen	Propulsion systems
	Steam	Boiler plant
	Light gases	Liquefaction
	Natural gas	Wellhead
350 bar (5000 lb/in ² g)	Drilling fluid (med)	Drilling
	Urea/carbonate	Urea production
	Hydrogen	Hydrogenation
	Synthesis gas	Ammonia
	Water	Oil field: reinjection
700 bar (10,000 lb/in ² g)	Methane	Oil field: reinjection
	Cement	Oil well 'kill'
	Water	Water-jet cutting
	Methane	Oil field: reinjection
	Natural gas	Wellhead
1400 bar (20,000 lb/in ² g)	Oil/water/gas (mixed)	Wellhead
	Oil/gas	Wellhead completion Wellhead choke and kill
2100 bar (30,000 lb/in ² g)	Ethylene	Low-density polyethylene production
10,000 bar (140,000 lb/in ² g)	Metal	Hydrostatic extrusion
50,000 bar (700,000 lb/in ² g)	Minerals	Synthetic diamond production

In fluid handling, certain processes require the use of high-pressure valves, the most widely known being ammonia synthesis, the oxy process, and processes for the production of urea and methanol as well as polymerisation processes for the production of low-density polyethylene. Apart from these, high-pressure valves are also required in hydrogenation processes, e.g. coal liquefaction or gasification.

The demands on the different items of process equipment vary depending on the process, operating pressure, operating temperature, and corrosivity of the media. Handling aggressive coal slurry, for example, which contains up to 30% pulverised coal at 10–30 microns particulate size can lead to enormous erosion problems in valves. Erosion can also be a problem in other types of high-pressure valves handling clean fluids, resulting from high localised fluid velocities. Flow-passage design, as well as material selection, is thus important in high-pressure valves. Such valves are, therefore, normally individual designs, not modifications of standard valve types with stronger components and greater wall thicknesses.

Special materials such as silicon nitride for seats and plugs may also be required to cope with severe conditions of abrasion and temperature. Special demand may also be placed on valve-stem seals, particularly as operating temperature may restrict material choice.

The ultimate test of a high-pressure valve is, however, the same as any other type of valve. It should be capable of performing its function reliably,



Hydraulic valve system for on/offshore applications.

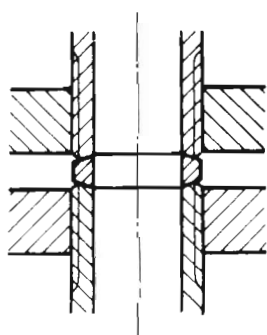
without leakage, and have an acceptable life. It is only the parameters that are more arduous.

Valve and pipe connections

In order to ensure sound leak-proof closures, all high-pressure joints should be precise in terms of concentricity, dimensional correctness, known material condition and predetermined bolt loading. Figures 1 to 5 show some examples.

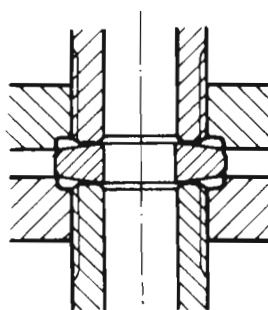
In order to generate the high closing forces required for larger valves at high pressure and to minimise operating time, a variety of actuators are used.

Hydraulic actuators (Figure 6) are ideal for larger valve sizes, although the hydraulic circuit and power pack can be expensive. Electrical actuators are



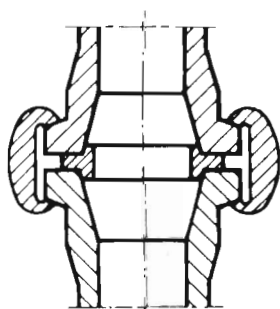
Lens ring
2000 to 3000 bar
1 inch and above

Figure 1.



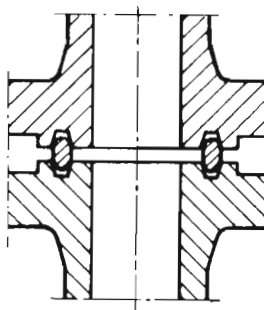
Cone ring
2000 to 3000 bar
Smaller sizes, *ie* instrument lines

Figure 2.



Hub and clamp seal
Up to 700 bar
All sizes

Figure 3.

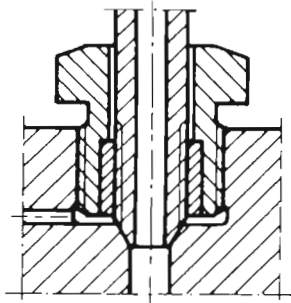


Ring joint
Up to 400 bar
All sizes

Figure 4.

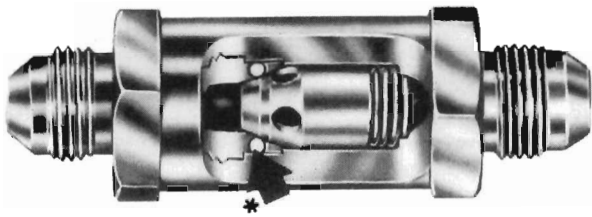
suitable for larger sizes provided torque is then transmitted through a suitable gearbox. Pneumatic actuators are suitable for smaller valves provided the mechanical advantage is increased via a suitable gearbox.

No matter what the application, it will be found that, in addition to high pressures, other conditions invariably prevail, whether it be temperature,



Union joint
2000 to 3000 bar
Small sizes

Figure 5.



High-pressure check valve rated to 6000 lb/in² with flexible seal seat.

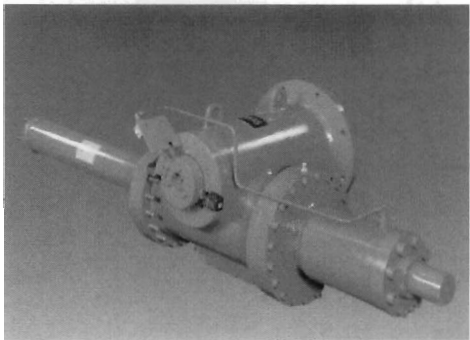
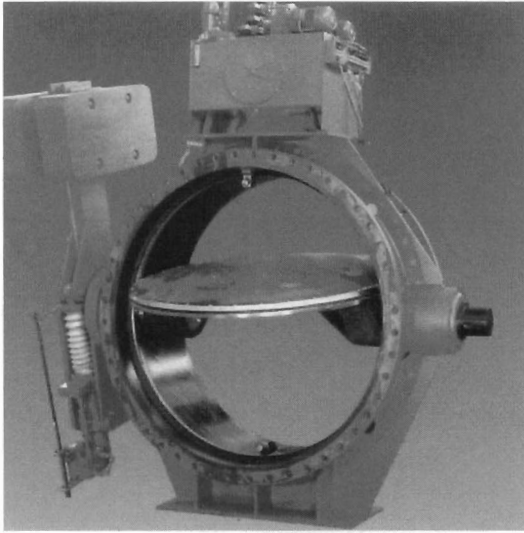


Figure 6. Hydraulic spring-return actuator.



Hydraulically-controlled, combined-function valve for hydro-electric power plant.

erosion, corrosion or a combination of either, resulting in the choice of materials being vital. Consultation with manufacturers would be advisable in all cases.

See also the chapters on Valve Actuators and Pipe Joints and Couplings.

For a more comprehensive coverage of high-pressure valves and applications, refer to the *Hydraulic Handbook*, also published by Elsevier Science Limited.

SECTION 9

Engineering Data

Glossary

Standards and Designations

Glossary

ABOVE GROUND: specifically referring to installations associated with a buried pipeline that are physically above the ground level (e.g. valves, etc.).

ABSOLUTE PRESSURE: pressure above absolute vacuum; or, in practice, gauge pressure plus atmospheric pressure.

ACME THREAD: square-cut thread form.

ACTUATOR: a device supplying force and motion to the closure member (ball, disc, plug, etc.) of a valve.

ADAPTOR: coupling or fitting used to connect pipes of different diameter sizes, join pipes of similar diameter but in different metals, join pipes with ends fabricated for different types of joints.

AERIAL MARKER: pipeline marker post visible from the air.

AMBIENT TEMPERATURE: temperature of the surrounding atmosphere.

ANCHOR BLOCK: concrete (or similar) block to which a pipe is attached, or in which it is embedded, to prevent movement.

ANGLE VALVE: type of globe valve with inlet and outlet passages at right angles, or some other angle (oblique valve).

ANNULAR FILL: material filling the annular space between a pipe and a sleeve surrounding it.

ANNULUS: annular space between a pipe and surrounding sleeve, or between two concentric pipes, etc.

ANODE: positive electrode in an electrolytic system (e.g. as in cathodic protection).

AQUEDUCT: man-made channel or pipe for carrying water.

BACKFILL: soil or other material used to fill in a trench.

BACK-PRESSURE: pressure acting against the outlet side of the valve.

BAG HOLE: hole cut in a pipe for insertion of a gas bag.

BAG PIPE: a hook or device for insertion or removal of an inflated gas bag from a gas main.

BALL COCK: valve whose movement is operated by a float attached to the end.

BALL VALVE: type of valve where the valve element is a spherical plug.

BAR: international standard unit of pressure.

BAR HOLE: small-diameter hole made by drilling or driving in a bar to search for source of leakage from an underground pipeline.